

A SaaS Multi-Vendor Ecommerce Platform

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ID# 21303039

A Practicum in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)



Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT—International University of Business Agriculture and Technology

Summer 2025

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The practicum has been examined and approved,

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Supervisor and Associate Professor

Department of Computer Science and Engineering
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IUBAT—International University of Business Agriculture and Technology

Summer 2025

Letter of Transmittal

20 September 2025

The Chair

Practicum Defense Committee

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Subject: Letter of Transmittal.

Dear Sir,

With due respect, I am pleased to submit my practicum report titled “A SaaS Multi-Vendor Ecommerce Platform” as part of the requirement for the B.Sc. in CSE at IUBAT. This study explores how to manage and maintain ecommerce website. I sincerely thank you and the Practicum Defense Committee for the opportunity and guidance. I have tried my best to ensure the accuracy and relevance of my practicum.

Your kind evaluation and valuable feedback will be truly appreciated.

Yours sincerely,

Md. Minhazul Islam

21303039

Organization's Certificate

I haven't received my certificate yet; hope I will get it tomorrow (21-09-2025.)

Student's Declaration

I hereby declare that the practicum titled “A SaaS Multi-Vendor Ecommerce Platform” is the result of my own work and has not been submitted previously for any degree or diploma in any institution.

I further declare that there is no plagiarism or data falsification in this report. All materials and information taken from various sources have been properly acknowledged and duly cited. I take full responsibility for the content and originality of this work.

Md. Minhazul Islam
21303039

Supervisor's Certification

This is to certify that Md. Minhazul Islam, ID# 21303039, has successfully completed the practicum titled “**A SaaS Multi-Vendor Ecommerce Platform**” under my supervision. The student's performance, work, and final report have been reviewed, and it is confirmed that the practicum requirements have been satisfactorily fulfilled. “Md. Minhazul Islam” is hereby recommended to appear for the practicum defense. Wishing the student continued success and excellence in all future academic and professional endeavors.

Drs Ferdaus Anam Jibon

Supervisor and Associate Professor

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Abstract

This project is about building a multi-vendor e-commerce platform where vendors can register, get verified, and manage their own online shops under one system. The idea is to provide a marketplace that works in a way similar to Shopify, where different vendors can sell products independently but still operate within a shared platform managed by an administrator.

The process begins with vendor registration. A vendor creates an account and uploads the necessary verification documents. This step is important because it ensures that only genuine vendors are allowed to sell on the platform. Once the admin reviews and approves the documents, the vendor is given the option to purchase a package or subscription plan. After buying a package, the vendor gains access to their shop management panel.

Inside the vendor panel, each vendor can set up their shop name, add products, manage categories, and create brands. Vendors also have tools to handle customer orders, making it easy to view order details, update order status, and track sales. This gives vendors complete control over how they want to display and manage their business online.

The admin panel works as the control center of the whole system. The administrator has the authority to approve vendors, manage the subscription packages, and monitor all activities on the platform. This balance of power allows vendors to run their shops freely while ensuring that the system remains safe, organized, and trustworthy.

The platform is designed to be scalable and flexible, so many vendors can join without affecting each other's business. At the same time, every vendor has the chance to build their own identity through their shop settings and branding. Secure authentication, document

verification, and role-based access are included to keep the platform safe for both vendors and customers.

In summary, this project aims to provide an easy and reliable online marketplace where vendors can start and grow their businesses without building a separate e-commerce site. It supports small and medium businesses by offering ready-made tools for online selling, while also giving administrators the ability to keep everything running smoothly.

Acknowledgments

First and foremost, I am deeply grateful to Almighty Allah for giving me the strength, patience, and ability to complete this practicum titled “A SaaS Multi-Vendor Ecommerce Platform.”

I would like to express my sincere appreciation to my honorable supervisor, Drs Ferdous Anam Jibon, for her valuable guidance, constructive feedback, and constant support throughout the practicum. I am also thankful to all the respected faculty members of the Department of Computer Science and Engineering, IUBAT, for their academic support and encouragement.

My heartfelt thanks go to my family and friends for their continuous motivation and inspiration. I also acknowledge the individuals and organizations who provided necessary data, insights, and technical support, which made this study possible.

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Chapter 1.
Introduction

This project is about creating a multi-vendor e-commerce platform where vendors can register, submit documents for verification, and, once approved by the admin, buy a package to get access to their own shop management panel. From there, vendors can manage products, categories, brands, and customer orders, as well as set up their shop name and identity.

The admin panel allows the administrator to approve vendors, control packages, and monitor overall activity, ensuring the system is safe and well managed. The platform is built to support many vendors at once, giving each the freedom to run their online shop while still keeping centralized control.

In short, the system provides a ready-to-use online marketplace that helps vendors grow their business without building a full e-commerce site on their own, while also making sure the marketplace stays secure, flexible, and easy to manage.

1.1 Background of the Study

Online shopping has become a normal part of people's daily lives, as it offers convenience and saves time compared to visiting physical stores. With the rise of digital commerce, more businesses are moving to online platforms to reach customers. However, while large companies can build and maintain their own e-commerce websites, many small and medium-sized vendors face challenges such as high costs, technical skills, and limited resources.

To overcome these issues, multi-vendor e-commerce platforms have become a practical solution. These platforms allow different vendors to create their own shops, manage products, and sell under one system. Examples like Shopify and Amazon Marketplace show how vendors can benefit from a shared marketplace while still building their brand identity.

The proposed project focuses on building such a system where vendors can register, submit documents for verification, and, after admin approval, buy a package to access their shop panel. From there, they can manage products, categories, brands, and orders. At the same time, the admin monitors vendor activities and manages subscriptions to ensure safety and fairness.

This platform aims to give vendors an easy way to start and grow their online business while offering customers a secure and reliable shopping experience.

1.2 Methodology

There are two types of data sources are used in this project. They are:

1.2.1 Primary Sources.

The steps of primary sources are given below:

- Interview with vendor, users and admin.
- Survey about multi-vendor ecommerce.
- Observation of current vendors selling products.

1.2.2 Secondary Sources.

The steps of secondary sources are given below:

- Online resources like website and existing system.
- Research Article and journal.

1.3 Objectives

The main objective of this project is to build a multi-vendor e-commerce platform where vendors can register, submit verification documents, and get approval from the administrator. After approval, vendors can purchase a package to access their shop management panel, allowing them to manage products, categories, brands, and customer orders. The platform also lets vendors create their shop identity and establish their brand. Administrators can monitor activities, approve vendors, and manage subscription plans to maintain a safe and organized system. The platform is designed to be user-friendly, flexible, and scalable,

enabling multiple vendors to operate efficiently while providing customers a smooth online shopping experience.

1.3.1 Broad Objective.

The broad objective of this project is to develop a multi-vendor e-commerce platform that allows vendors to register, get verified, and manage their shops, including products, categories, brands, and orders. Vendors can build their shop identity, while administrators oversee approvals and subscriptions. The system is designed to be secure, flexible, and user-friendly, supporting multiple vendors and enhancing the online shopping experience.

1.3.2 Specific Objectives.

- To develop a vendor registration and verification system that ensures only genuine vendors can access the platform.
- To create a shop management panel for vendors to manage products, categories, brands, and customer orders efficiently.
- To implement a subscription-based package system allowing vendors to unlock features based on their chosen plan.
- To design an administrator dashboard for approving vendors, monitoring activities, and managing subscriptions to maintain platform security and organization.

1.4 Process Model

The project is developed using the Agile Scrum process model. Scrum divides the project into short, time-boxed iterations called sprints, where each sprint delivers a working part of the system. This allows continuous development, testing, and feedback, ensuring that the system evolves according to user requirements. Regular meetings such as sprint planning, daily stand-ups, and sprint reviews help the team stay organized and maintain transparency throughout the development process. Reason for choosing this model are given below-

- **Flexibility:** Requirements may change during development, and Scrum adapts easily to new needs.
- **Incremental Delivery:** Working features are delivered in every sprint, providing early usable versions.
- **User Involvement:** Continuous feedback from stakeholders ensures the system matches real-world needs.

- **Collaboration:** Encourages teamwork, communication, and shared responsibility among developers.
- **Risk Reduction:** Early testing in each sprint helps detect and solve issues quickly.
- **Higher Quality:** Regular reviews and improvements ensure a more reliable and user-focused product.

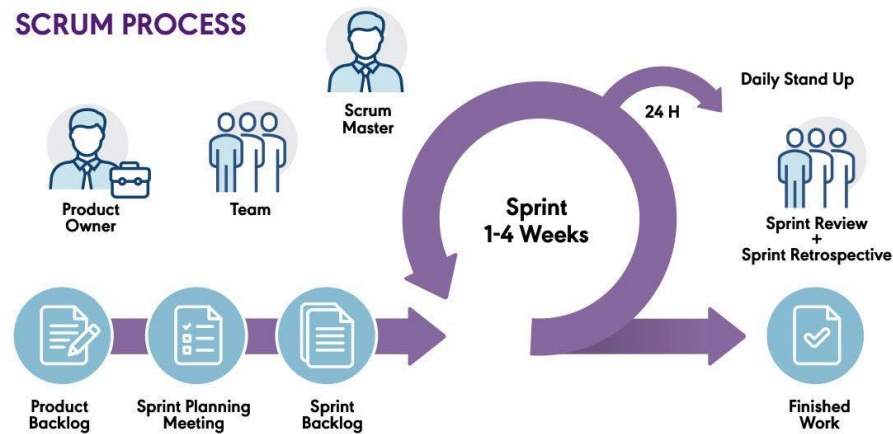


Figure 1.4.1 Agile Scrum Process Model

1.5 Feasibility Study

The feasibility study has been divided into three phases such as technical feasibility, economic feasibility, and operational feasibility. The proposed system is feasible in all these three phases as discussed below:

1.5.1 Technical Feasibility.

The proposed multi-vendor e-commerce platform is technically feasible because the required tools, technologies, and resources are readily available and widely used in web application development. The system can be built using modern web technologies such as HTML, CSS, JavaScript, and frameworks like Bootstrap for the front end to ensure a user-friendly interface. For the back end, PHP with MySQL and framework laravel can be used to handle vendor registration, product management, order processing, and secure data storage. These technologies are open-source, cost-effective, and supported by large developer communities.

File upload and verification for vendor documents can be managed with existing libraries and cloud storage solutions. Furthermore, subscription-based features can be integrated using payment gateways such as SSLCommerz, which are reliable and easy to implement.

Since these tools are well-documented and compatible, the development team can design a scalable and flexible platform without the need for highly specialized hardware or expensive proprietary software. Thus, the system is technically feasible, achievable within available resources, and sustainable for future upgrades.

1.5.2 Economic Feasibility.

The proposed multi-vendor e-commerce platform is economically feasible because it can be developed and maintained with affordable resources compared to its expected benefits. The project will rely mostly on open-source technologies such as PHP, MySQL, and Laravel, which do not require costly licenses. This reduces development costs and makes the platform more sustainable in the long term. Hosting and storage services are also available at low cost

through cloud providers, ensuring flexibility and scalability without heavy investment in physical infrastructure.

From a revenue perspective, the system has strong earning potential. Vendors will pay for subscription packages to access the shop management panel, and additional income can be generated through transaction fees, advertisements, or premium features. These earnings can quickly cover the initial development costs and provide ongoing profit.

Since the investment mainly involves development time, basic infrastructure, and minimal recurring expenses, the project ensures a favorable cost-benefit ratio. The financial return from vendor subscriptions and marketplace growth is expected to outweigh the setup and maintenance costs, making the project economically sustainable.

1.5.3 Operational Feasibility.

The proposed multi-vendor e-commerce platform is operationally feasible because it is designed to meet the needs of administrators, vendors, and customers in a simple and organized way. The system ensures smooth onboarding by allowing vendors to register and submit documents for verification, while administrators can easily approve or reject applications from their dashboard. Once approved, vendors can purchase a package and access their shop panel, where they can manage products, categories, brands, and customer orders with minimal training.

For administrators, the platform offers centralized control to monitor vendor activities, manage subscriptions, and ensure the marketplace runs securely. Customers will also benefit from a user-friendly interface that makes it easy to browse products, place orders, and track deliveries.

Since all users are given role-based access, the system avoids confusion and ensures that each group interacts only with the tools they need. The platform's design is straightforward, requiring little technical knowledge from vendors and customers, making it highly practical for everyday use. With proper training for administrators and clear guidance for vendors, the system can operate efficiently and fulfill its intended purpose.

1.6 Structure of the Report

The report is structured into ten chapters, beginning with this Introduction, followed by Organizational Overview, Requirement Engineering, Analysis Modeling, Project Management, Project Planning and Scheduling, Project Estimation, Designing, Testing, and Conclusion, and ending with the References.

Chapter 2.

Organizational Overview

SSTechnoSys Ltd. is a leading technology solutions provider specializing in digital transformation. The company offers expertise in software development, IT consultancy, and enterprise automation, delivering intelligent, scalable, and future-ready solutions tailored to the unique needs of clients.

Founded with the vision to reshape the IT landscape, SSTechnoSys combines cutting-edge technology with strategic insight to help businesses thrive in a rapidly evolving digital world. Over the years, the company has built a strong reputation for excellence, client-centric services, and innovative problem-solving, ensuring customized solutions for complex business challenges.

2.1 Organization Vision

To empower businesses with next-generation technology solutions that enhance efficiency, security, and growth.

2.2 Organization Mission

To empower businesses with next-generation technology solutions that enhance efficiency, security, and growth.

2.3 Organization Services

Provide different IT Solutions for Successful Business. Some of them are-

- **Industrial and Office Automation:** Provide cutting-edge automation solutions that streamline operations, reduce costs, and enhance efficiency in both industrial and corporate environments.
- **Custom Software Development:** We specialize in custom software solutions tailored to the specific needs of businesses.
- **System Support and IT Infrastructure:** We offer comprehensive IT support services to ensure smooth and secure business operations.

2.4 Organizational Structure

The organogram of the organization has shown in figure 2.4.

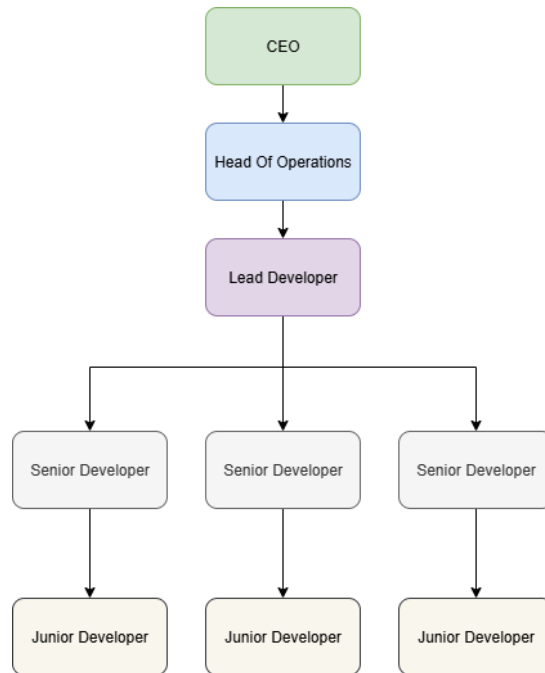


Figure 2.4 Organizational Structure

2.5 My Position in this Organization

As an Intern Web Developer at SSTechnoSys Ltd., I will support the design, development, and maintenance of web applications that align with client and business needs. Working under the guidance of senior developers, I will gain hands-on experience in front-end and back-end technologies, coding best practices, and collaborative software development. This role provides a valuable opportunity to enhance my technical skills, contribute to real-world projects, and understand the workflow of a professional IT solutions environment.

2.6 Address of the Organization

House#17, Road#11, Sector#10, Uttara, Dhaka.

Chapter 3.

Requirement Engineering

Requirement Engineering (RE) is the systematic process of defining, documenting, and maintaining the requirements for a software system. It ensures that the software being developed meets the needs of stakeholders (clients, users, and businesses) and provides a clear foundation for design, development, and testing. There are mainly three steps of requirement engineering-

- **Requirement Elicitation:** Gathering requirements from stakeholders through interviews, questionnaires, workshops, observations, etc.
- **Requirement Analysis:** Examining, refining, and prioritizing the collected requirements. Detecting conflicts, inconsistencies, or feasibility issues.
- **Requirement Specification:** Documenting requirements in a structured form, usually in a Software Requirement Specification (SRS) document

Detailed explanation of these is given below for my project A SasS Multi-Vendor Ecommerce.

3.1 Requirement Elicitation

In Requirement Elicitation, we gather information from stakeholders such as clients, users, and managers to understand what they need from the software. This involves asking questions, conducting interviews or surveys, and observing existing processes to identify problems and expectations. The goal is to collect clear, complete, and accurate information so that the system can be designed to meet the actual needs of its users.

The following task is completed for my project.

3.1.1. Requirement Gathering

I conducted direct interviews with stakeholders, including vendors interested in developing an e-commerce platform, to collect detailed information about their needs, expectations, and functional requirements for the system.

3.2 Requirement Analysis

In this step we analyze the requirement and find out specific features from user stories. We filter requirement according to our needs. In this step we just find out primary requirement from user stories. Primary requirement is given below:

- **Vendor Requirement:**
 - Manage product category.
 - Manage product brands.
 - Manage product attributes.
 - Manage all products including crud.
 - Manage orders of all products.
 - Buy and see their package they purchased.
 - See their profile details.
 - Setup basic shop information.
- **User Requirement:**
 - User can browse vendor shop.
 - User can order product of different vendor.
 - User can see their order list based on vendor.
- **Admin Requirement:**
 - Admin can manage all verification request.
 - Admin can manage package (add, edit, delete).
 - Admin can monitor all vendors, users and overall system.

Additional requirements expected by client.

- User friendly environment with attractive simple UI.
- Different dashboard for Admin, Vendor and User.
- Secure login and registration for every stakeholder.
- Notify user when order using email.

3.3 Requirement Specifications

In this step we documented detailed requirements for our system. Both functional and non-functional requirements are included here.

3.3.1. User Requirements

1. Vendor can register on the platform and submit verification documents.
2. Vendor can access a shop management panel after admin approval and package purchase.
3. Vendor can add, edit, and delete products.
4. Vendor can create, edit, and manage product categories.
5. Vendor can create, edit, and manage brands.
6. Vendor can manage customer orders, update order status, and track sales.
7. Vendor can set up and customize their shop name and branding.
8. Vendor can securely log in and manage their account information.
9. Administrator can approve or reject vendor registrations.
10. Administrator can manage subscription packages and assign access rights.
11. Administrator can monitor all vendor activities and orders.
12. Customer can browse products from multiple vendors using their shop URL.

13. Customer can place orders and make payments securely.
14. Customer can track order status and delivery.
15. Customer can create and manage a personal account.

3.3.2. System Requirements

1. Vendor Registration and Document Submission

- 1.1. Vendor press gets started button go to registration page.
- 1.2. Vendor will give email and password for registration and click register button.
- 1.3. After successful registration it will show success message.
- 1.4. Vendor click login button which will open login page.
- 1.5. Vendor give their registered email and password and click login page which will land them to platform dashboard.
- 1.6. User click in verification from left sidebar which will show form for submit required documents.
- 1.7. User upload all documents including NID, Trade License, TIN certificate and press Submit button.
- 1.8. After successful submission it will show success message.

2. Package Purchase and activation

- 2.1. Click add new service from left-side bar which will show all available packages.
- 2.2. Choose desired package and click buy button which will land in review page.
- 2.3. Select payment method and click confirm button which will land in payment page. After completing payment active packages will show in status page.
- 2.4. From status page click active button which will make package active.
- 2.5. After activating and verifying documents click go to shop admin panel button which will land shop management admin panel.

3. Product Management

- 3.1. Click add new product button which will show product adding form.
- 3.2. Input information including title, slug, price, description, images, category, stock, discount, status, feature and brand.
- 3.3. Click Save to add the product to the shop.
- 3.4. Select a product from the list and click **Edit** to modify product information.
- 3.5. Select a product from the list and click Delete to remove it from the shop.

4. Category Management

4.1. Click Add New Category button to open the category form.

4.2. Input category name, slug, icon, image, feature and status.

4.3. Click Submit to create a new category.

4.4. Select a category from the list and click Edit to modify it.

4.5. Select a category from the list and click Delete to remove it.

5. Brand Management

5.1. Click Add New Brand button to open the brand form.

5.2. Input brand name, slug, and status.

5.3. Click Submit to create a new brand.

5.4. Select a brand from the list and click Edit to modify it.

5.5. Select a brand from the list and click Delete to remove it.

6. Order Management

6.1. Click Orders in the vendor dashboard to view all customer orders.

6.2. Click an order to see order details including customer information, products, and total amount.

6.3. Update order status (e.g., Pending, Processing, Shipped, Delivered).

6.4. Track sales statistics from the dashboard reports.

7. Shop Customization

7.1. Click Shop Settings in the vendor dashboard.

7.2. Input shop name, description, and vendor name

7.3. Click Update to apply changes to the shop appearance.

8. Vendor Account Management

8.1. Click Profile in the dashboard.

8.2. Update personal information such as name, email, phone, and password.

8.3. Click Submit to apply changes.

9. Vendor Registration Approval (Admin)

9.1. Admin clicks Pending Vendors in the admin panel.

9.2. View vendor registration details and submitted verification documents.

9.3. Click Approve to grant access to shop management after package purchase.

9.4. Click Reject to decline registration with a reason.

10. Subscription Management (Admin)

- 10.1. Admin clicks Subscription Packages in the admin panel.
- 10.2. Create, edit, or delete packages with details such as name, price, and features.

11. Vendor Activity Monitoring (Admin)

- 11.1. Admin clicks Vendor Dashboard to view all registered vendors.
- 11.2. Monitor vendor activities, product listings, and order updates.
- 11.3. Generate reports on sales, orders, and vendor performance.

12. Customer Product Browsing

- 12.1. Customers visit shop URL to view products.
- 12.2. Use search bar to find products by keyword.
- 12.3. Apply filters by category, brand, or price range.
- 12.4. Click a product to view product details including description, price, and vendor.

13. Customer Order & Payment.

- 13.1. Click Add to Cart to select products for purchase.
- 13.2. Proceed to Checkout and fill in shipping information.

13.3. Choose payment method and complete secure payment.

13.4. Receive order confirmation and invoice.

14. Customer Order Tracking

14.1. Click Orders in customer account.

14.2. View order status (e.g., Pending, Processing, Shipped, Delivered).

15. Customer Account Management

15.1. Click My Profile in account dashboard.

15.2. Update personal information including name, email, phone, and password.

15.3. Save changes to maintain an up-to-date account.

3.3.3. Functional Requirements

1. Vendor Registration

- The system shall allow vendors to register using email and password.
- The system shall display a success message upon successful registration.
- The system shall allow vendors to log in with registered credentials.

2. Vendor Verification

- The system shall allow vendors to submit verification documents (NID, Trade License, TIN certificate).
- The system shall display a confirmation message after successful document submission.
- The system shall enable administrators to approve or reject vendor verification.

3. Package Purchase and Activation

- The system shall allow vendors to view available subscription packages.
- The system shall allow vendors to purchase a package through a secure payment gateway.
- The system shall activate the purchased package and update its status.
- The system shall allow vendors to access the shop management panel after verification and package activation.

4. Product Management

- The system shall allow vendors to add new products with details such as title, description, price, images, stock, category, brand, discount, and status.
- The system shall allow vendors to edit existing product information.

- The system shall allow vendors to delete products.

5. Category Management

- The system shall allow vendors to create, edit, and delete product categories.
- The system shall allow vendors to assign products to categories.

6. Brand Management

- The system shall allow vendors to create, edit, and delete brands.
- The system shall allow vendors to assign products to brands.

7. Order Management

- The system shall allow vendors to view all customer orders.
- The system shall display detailed order information (products, quantity, customer info, total amount).
- The system shall allow vendors to update order status (Pending, Processing, Shipped, Delivered).
- The system shall provide sales reports and order statistics.

8. Shop Customization

- The system shall allow vendors to customize shop name, description, and branding.

- The system shall save and display shop customization changes in the vendor shop.

9. Vendor Account Management

- The system shall allow vendors to view and update personal information (name, email, phone, password).

10. Vendor Registration Approval (Admin)

- The system shall allow administrators to view pending vendor registrations.
- The system shall display submitted verification documents.
- The system shall allow administrators to approve or reject vendors with a reason.

11. Subscription Management (Admin)

- The system shall allow administrators to create, edit, or delete subscription packages.
- The system shall allow administrators to manage package features and prices.

12. Vendor Activity Monitoring (Admin)

- The system shall allow administrators to view all registered vendors.

- The system shall allow administrators to monitor vendor activities, products, and orders.
- The system shall generate reports on sales, orders, and vendor performance.

13. Product Browsing (Customer)

- The system shall allow customers to view products from multiple vendors.
- The system shall provide search functionality by keyword.
- The system shall allow filtering by category, brand, or price.

14. Order Placement and Payment (Customer)

- The system shall allow customers to add products to the cart.
- The system shall allow customers to provide shipping information and complete secure payment.
- The system shall generate order confirmation and invoice.

15. Order Tracking (Customer)

- The system shall allow customers to view the status of their orders (Pending, Processing, Shipped, Delivered).

16. Customer Account Management

- The system shall allow customers to create and manage personal accounts.
- The system shall allow customers to update personal information (name, email, phone, password).

3.3.4. Non-Functional Requirements

1. Security

- The system shall use HTTPS/SSL for all communications.
- Passwords shall be stored using secure hashing algorithms (e.g., bcrypt).
- Role-based access control (vendor, admin, customer) shall restrict access to sensitive functions.
- Uploaded documents (NID, Trade License, TIN) shall be validated and stored securely.
- Payments shall be processed through secure, PCI-compliant payment gateways.

2. Usability

- The platform shall have a responsive interface for desktop, tablet, and mobile users.

- Users shall receive clear feedback messages for actions like login, registration, payment, or errors.
- Product search, filter, and sorting shall be intuitive and easy to use.
- Vendor and admin dashboards shall be easy to navigate, with well-organized menus.

3. Scalability

- The system shall support adding more vendors, products, and customers without downtime.
- The architecture shall allow easy addition of new features, such as promotions, analytics, or multi-language support.

4. Maintainability

- The system shall be built using modular code architecture for easy updates and bug fixes.
- Logging and monitoring tools shall track system errors, performance issues, and user activities.
- Admins shall be able to update subscription packages, categories, and brands without technical assistance.

3.4 Use Case Diagram

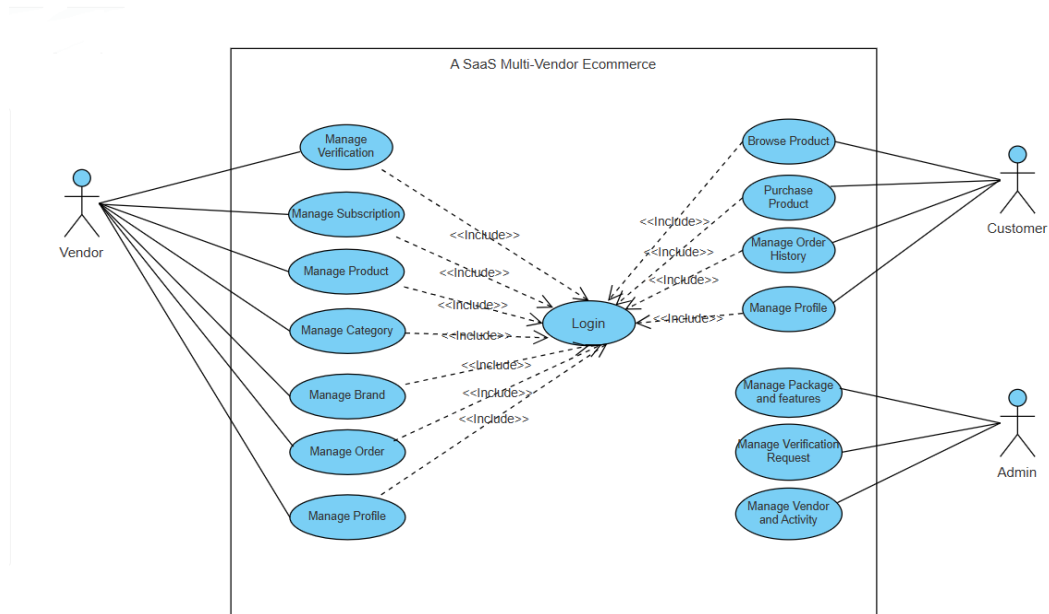


Figure 3.4 Use Case Diagram.

Chapter 4.

Analysis Modeling

In this chapter I will illustrate various diagram related to my project manager including activity diagram, swim lane diagram, sequence diagrams, CRC and class diagrams.

4.1 Activity Diagram

An activity diagram is a visual representation used in software engineering to describe the sequence of actions or steps within a process. It illustrates how activities flow from one to another, including decisions, parallel actions, and possible outcomes. This type of diagram is commonly used to understand, analyze, and design workflows or system behaviors, making it easier to identify the logic of a process and improve communication between developers, stakeholders, and users.

4.1.1. Vendor Registration

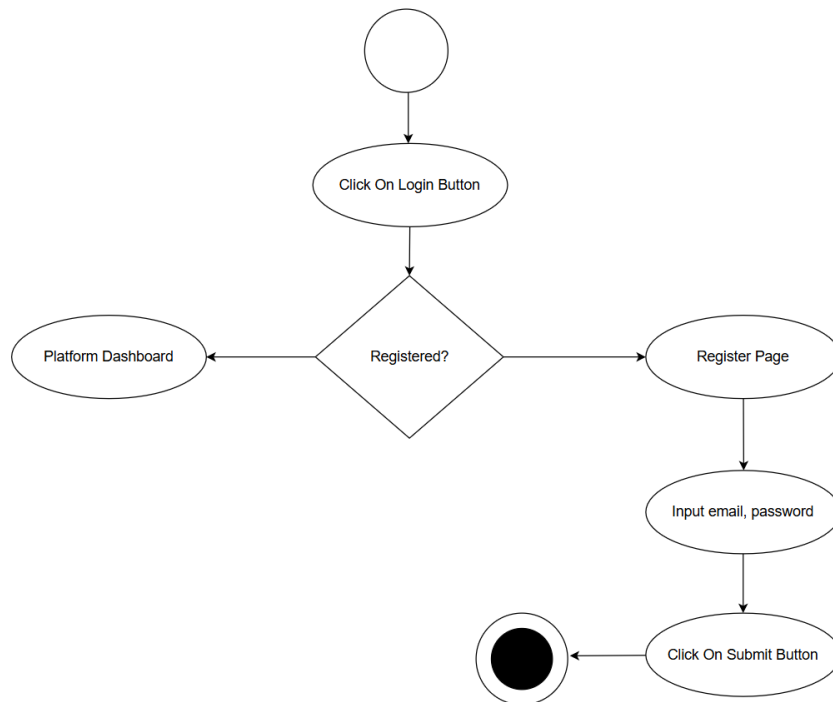


Figure 4.1.1 Vendor registration activity diagram

4.1.2. Document Upload

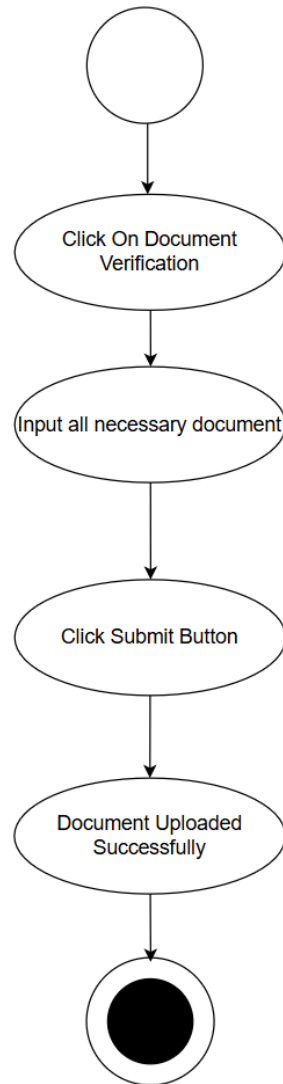


Figure 4.1.2 Document Upload activity diagram

4.1.3. Purchase Package

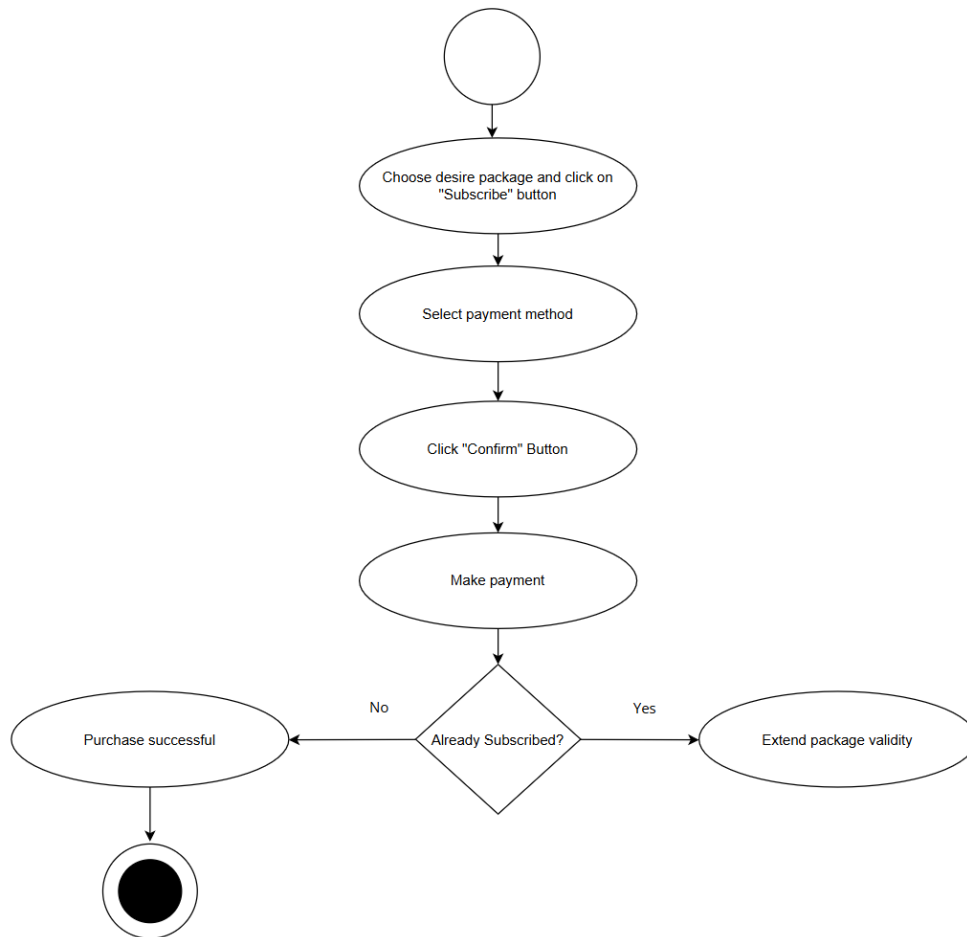


Figure 4.1.3 Purchase Package.

4.1.4. Package Activation

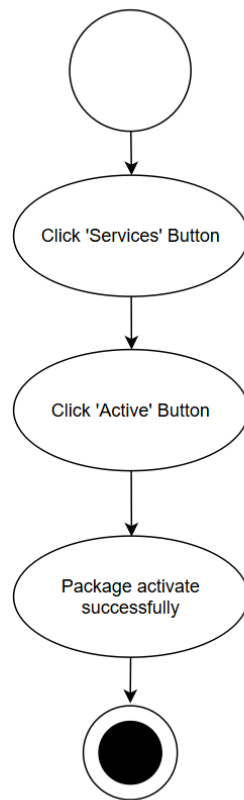


Figure 4.1.4 Package Activation.

4.1.5. Vendor Approval

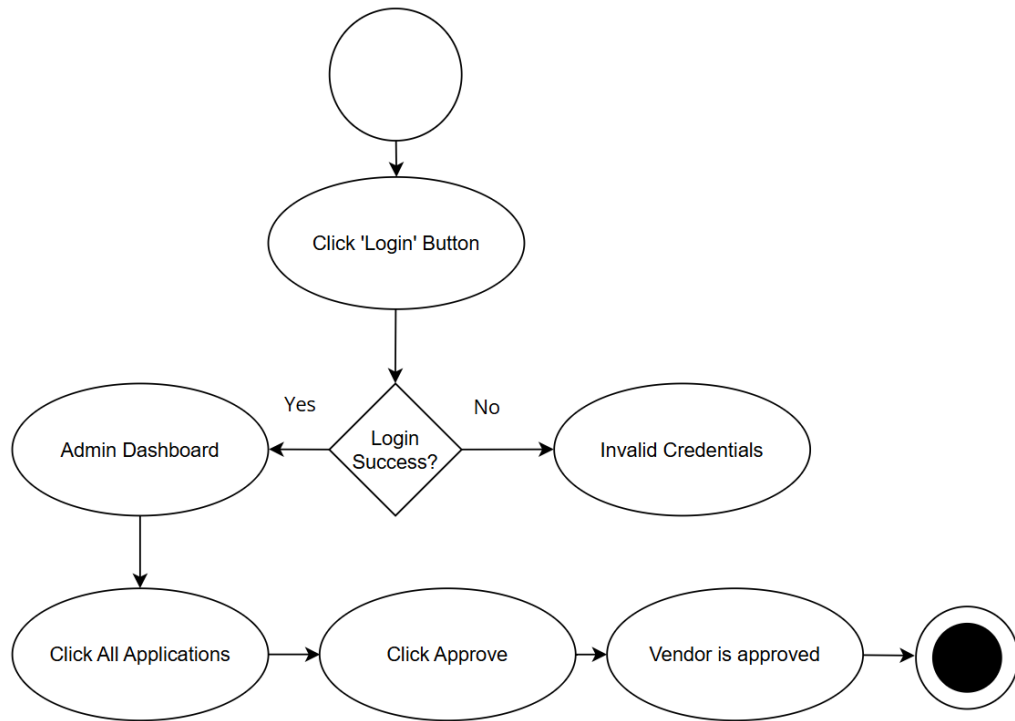


Figure 4.1.5 Vendor Approval.

4.1.6. Add New Product

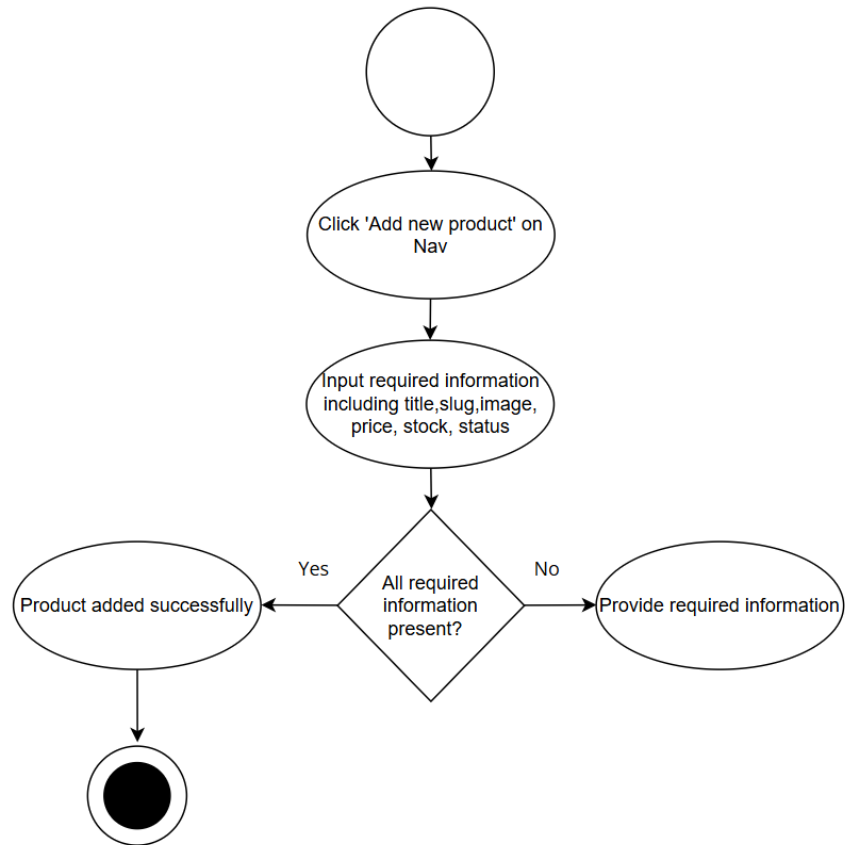


Figure 4.1.6 Add new product.

4.1.7. Product Order Process

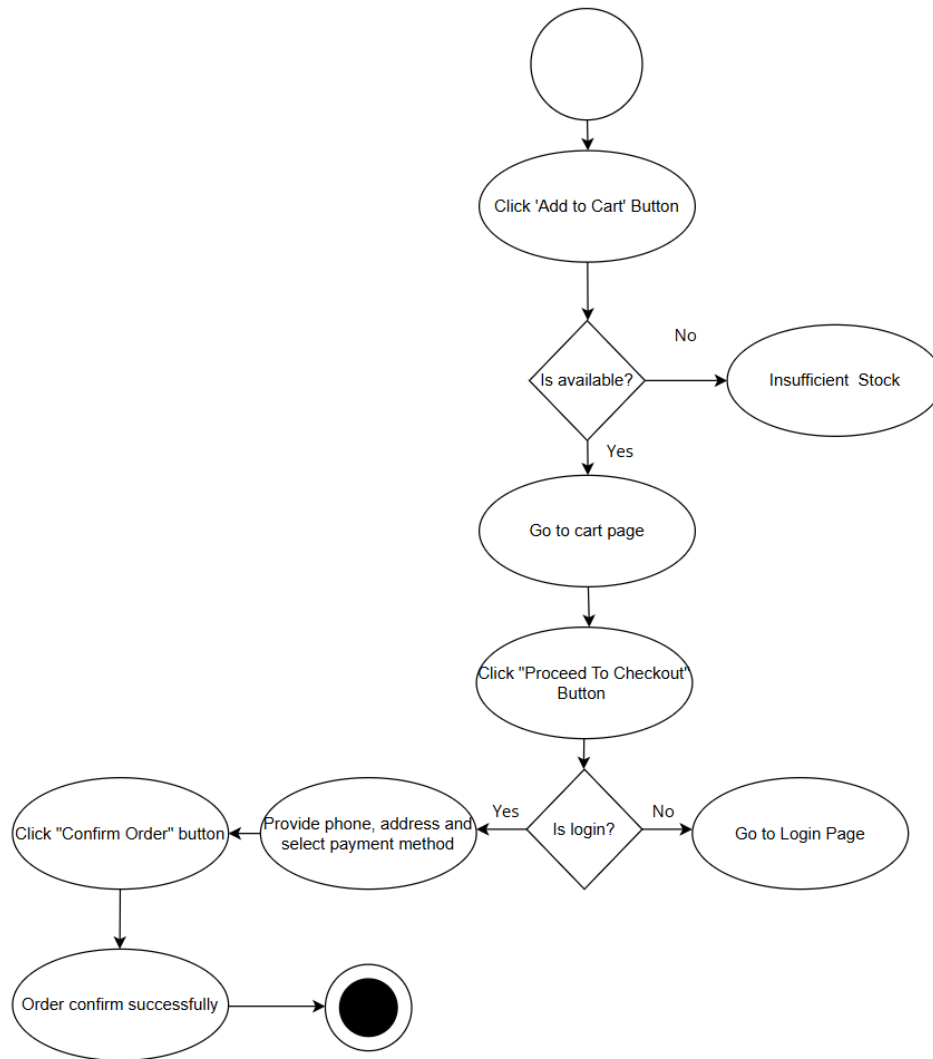


Figure 4.1.7 Product Order System.

4.1.8. Order Management Process

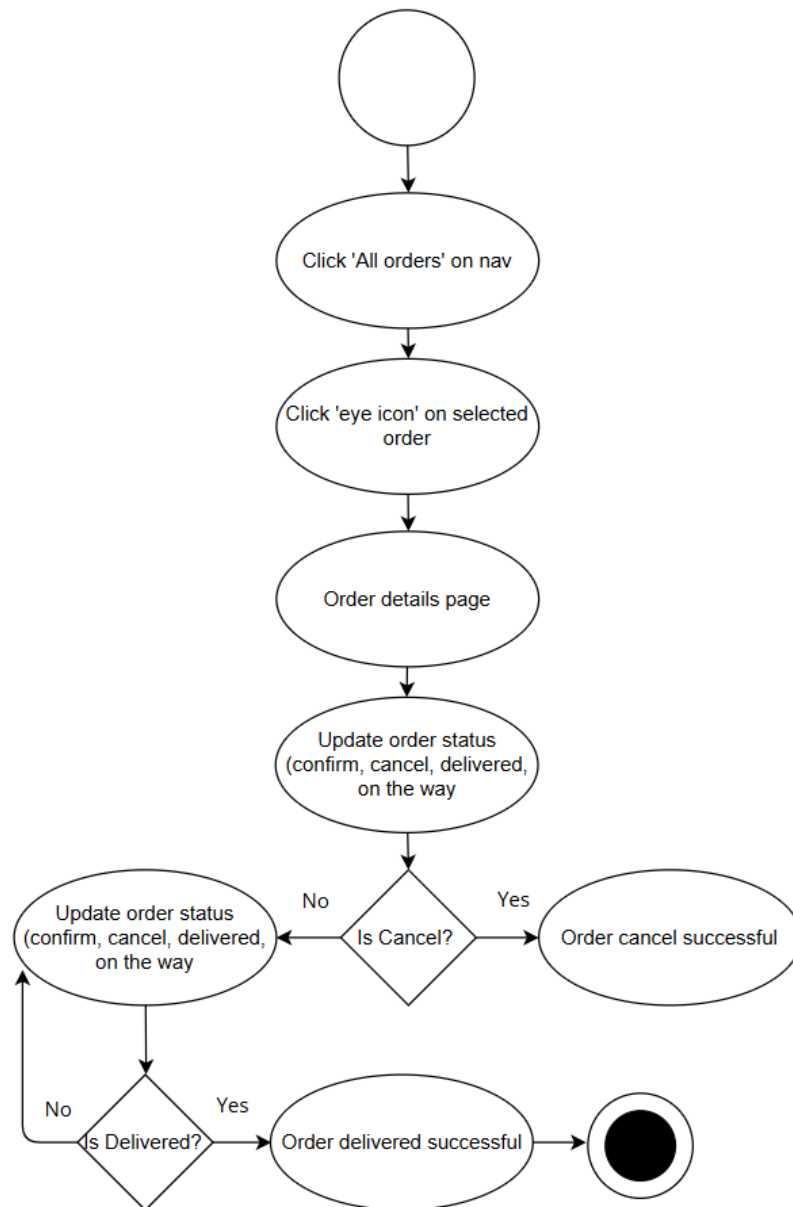


Figure 4.1.8 Order Management Process.

4.2 Swimlane Diagram

A swimlane diagram is a type of flowchart used to represent a process by dividing it into lanes, where each lane corresponds to a role, department, or system. It visually organizes tasks and responsibilities, showing how different participants interact and how work flows across them. This makes it easier to understand who is responsible for each step, identify handoffs, and detect inefficiencies or gaps in a process.

4.2.1. Vendor Registration

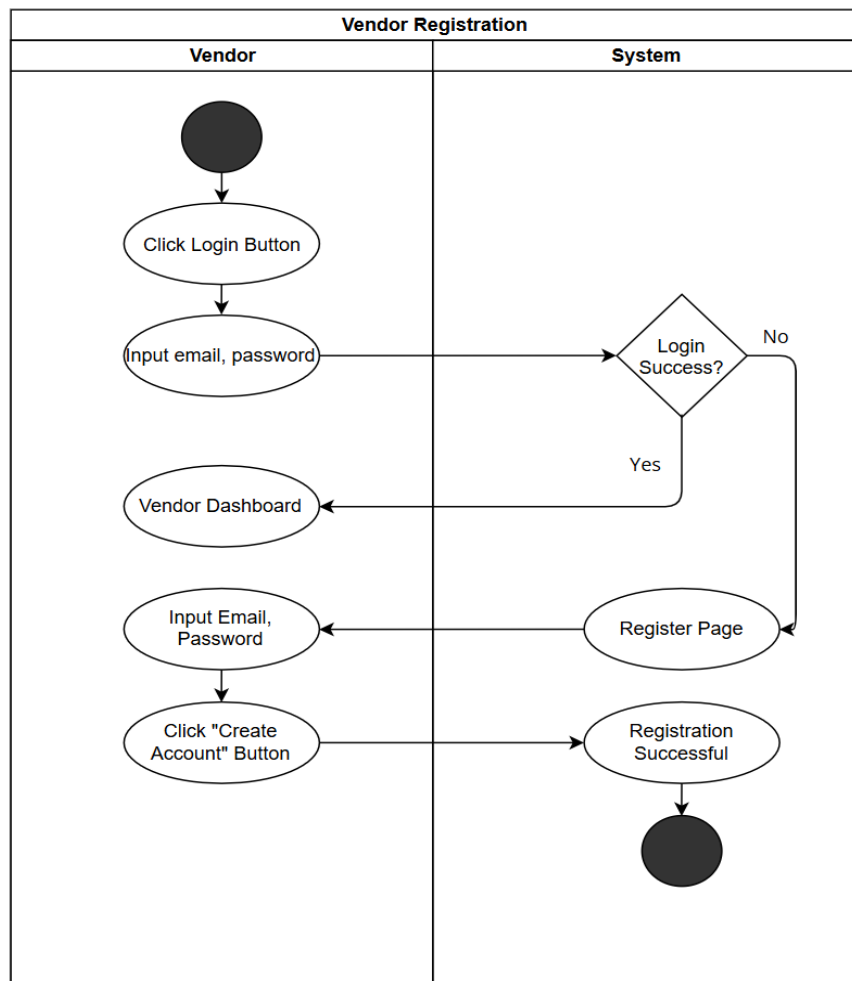


Figure 4.2.1 Vendor Registration.

4.2.2. Document Upload

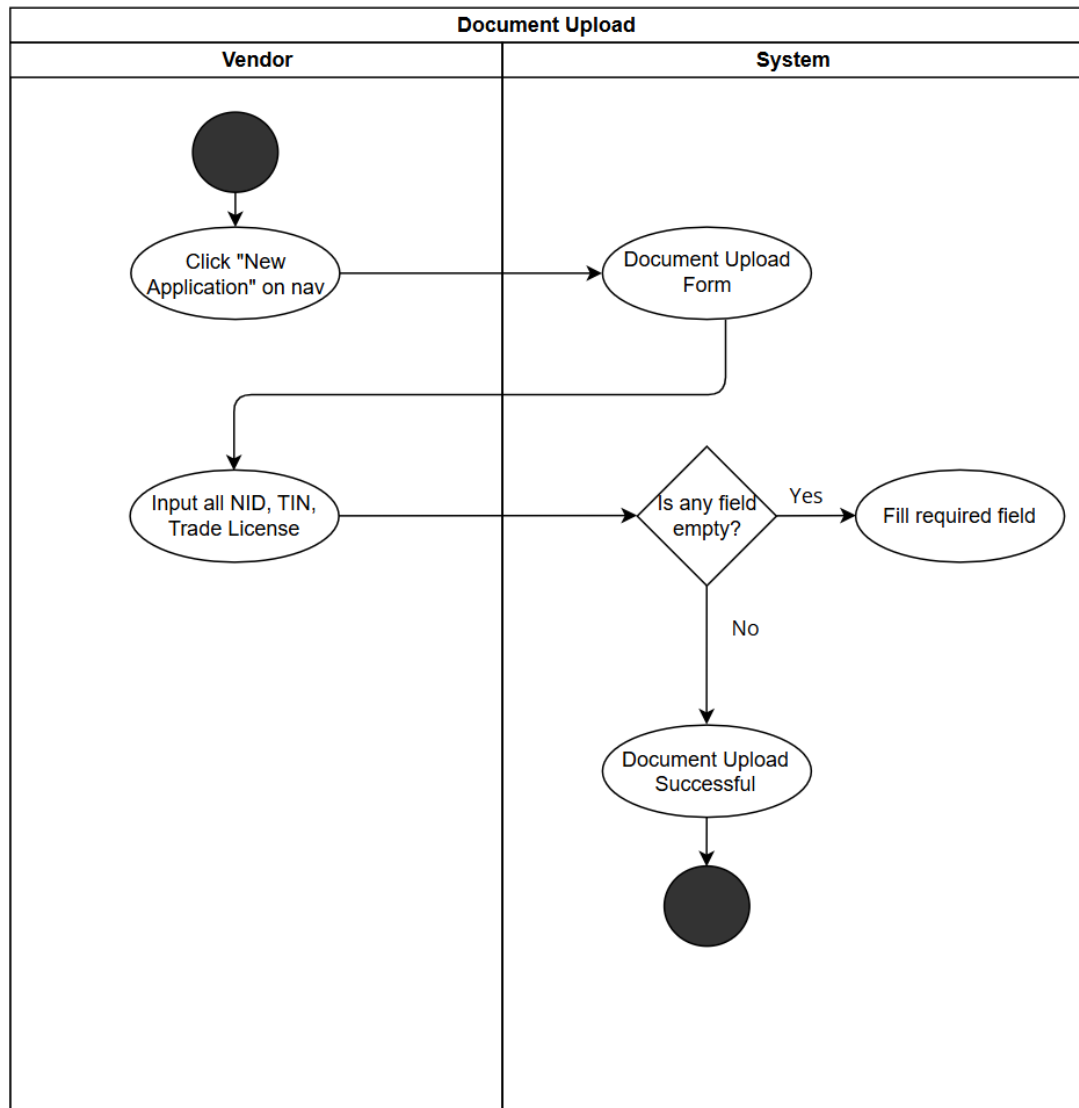


Figure 4.2.2 Document Upload

4.2.3. Package Purchase

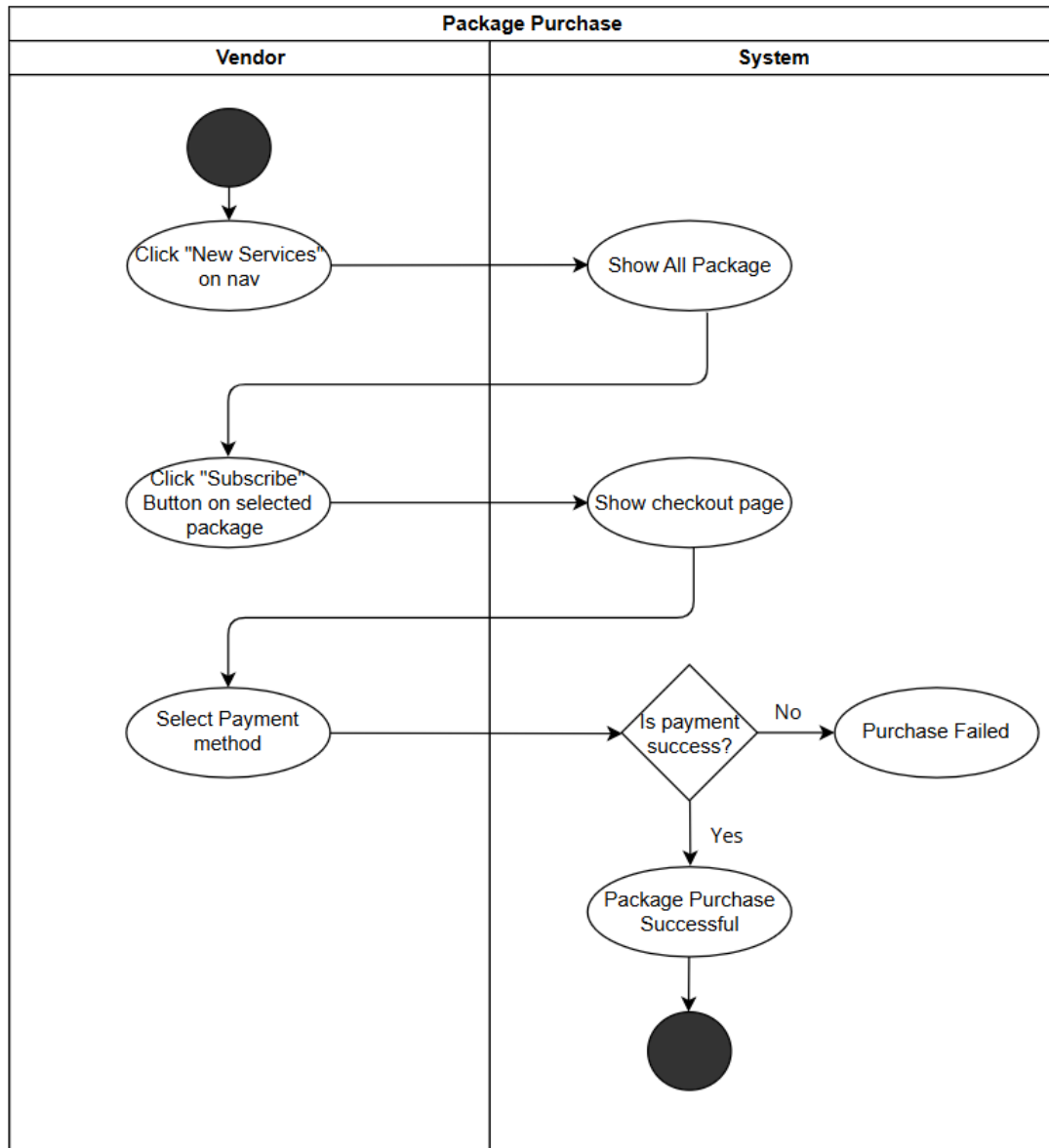


Figure 4.2.3 Package Purchase.

4.2.4. Package Activation

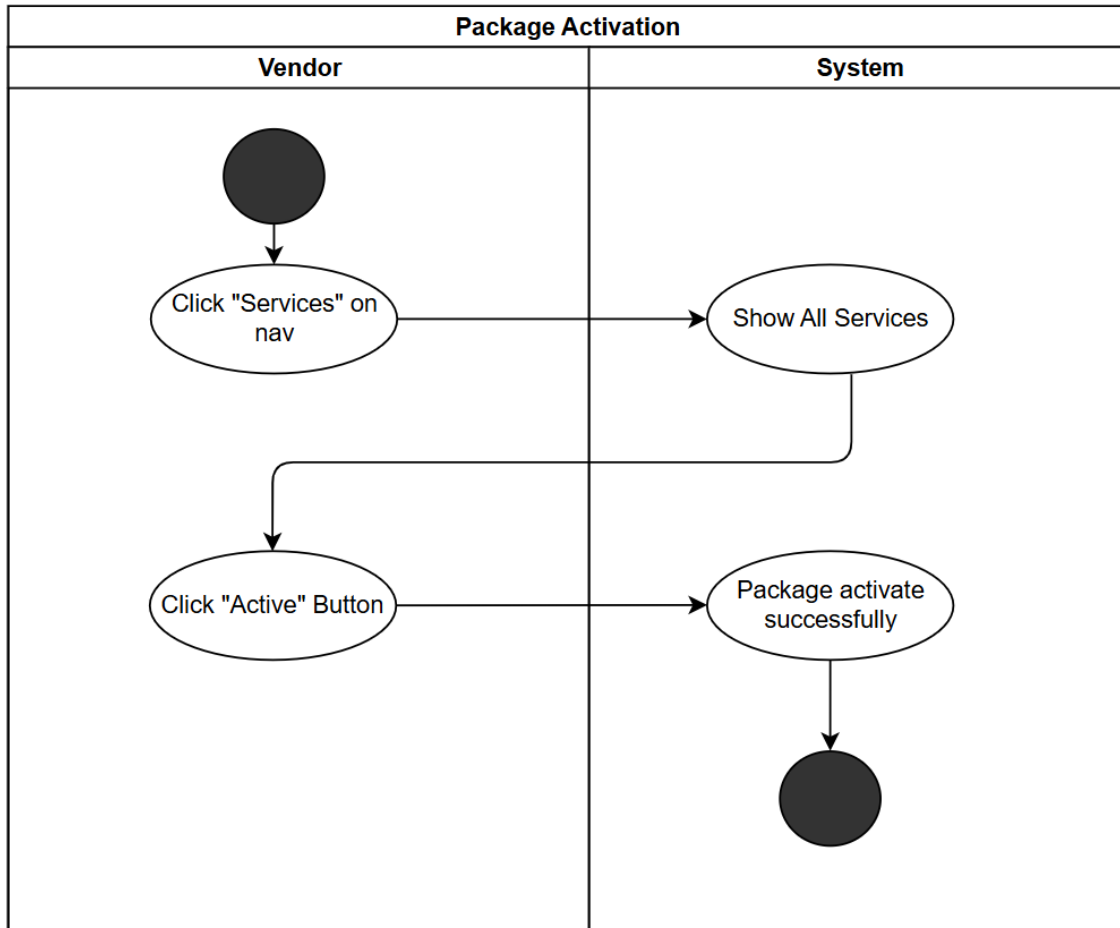


Figure 4.2.4 Package Activation.

4.2.5. Vendor Approval

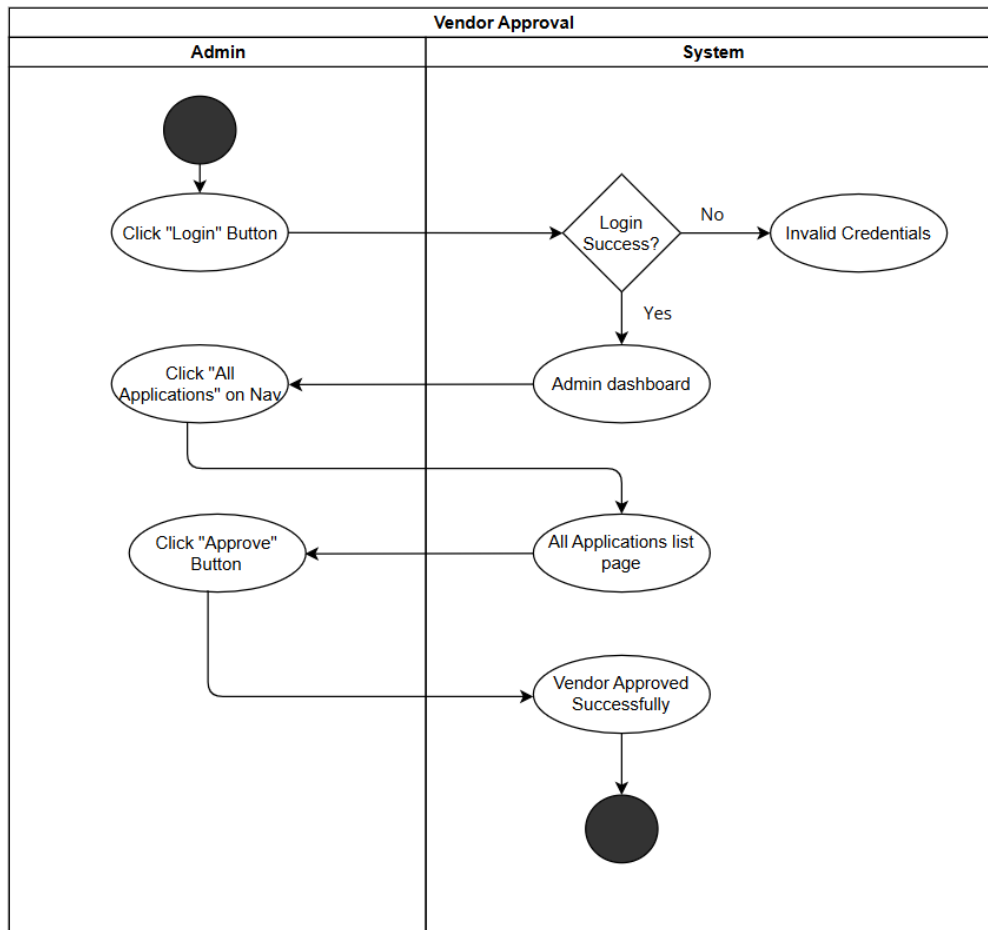


Figure 4.2.5 Vendor Approval.

4.2.6. Add New Product

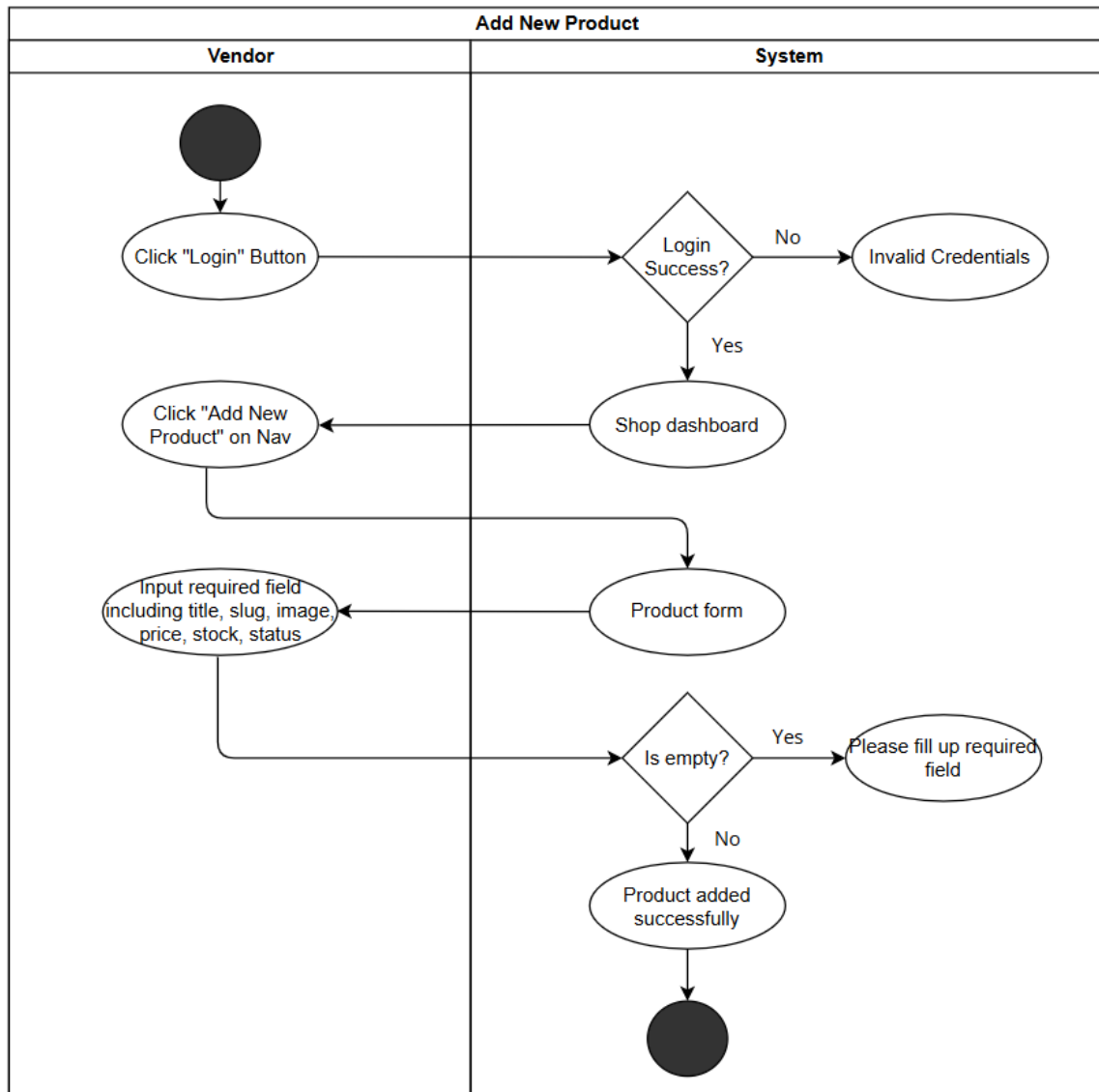


Figure 4.2.6 Add New Product.

4.2.7. Product Order Process

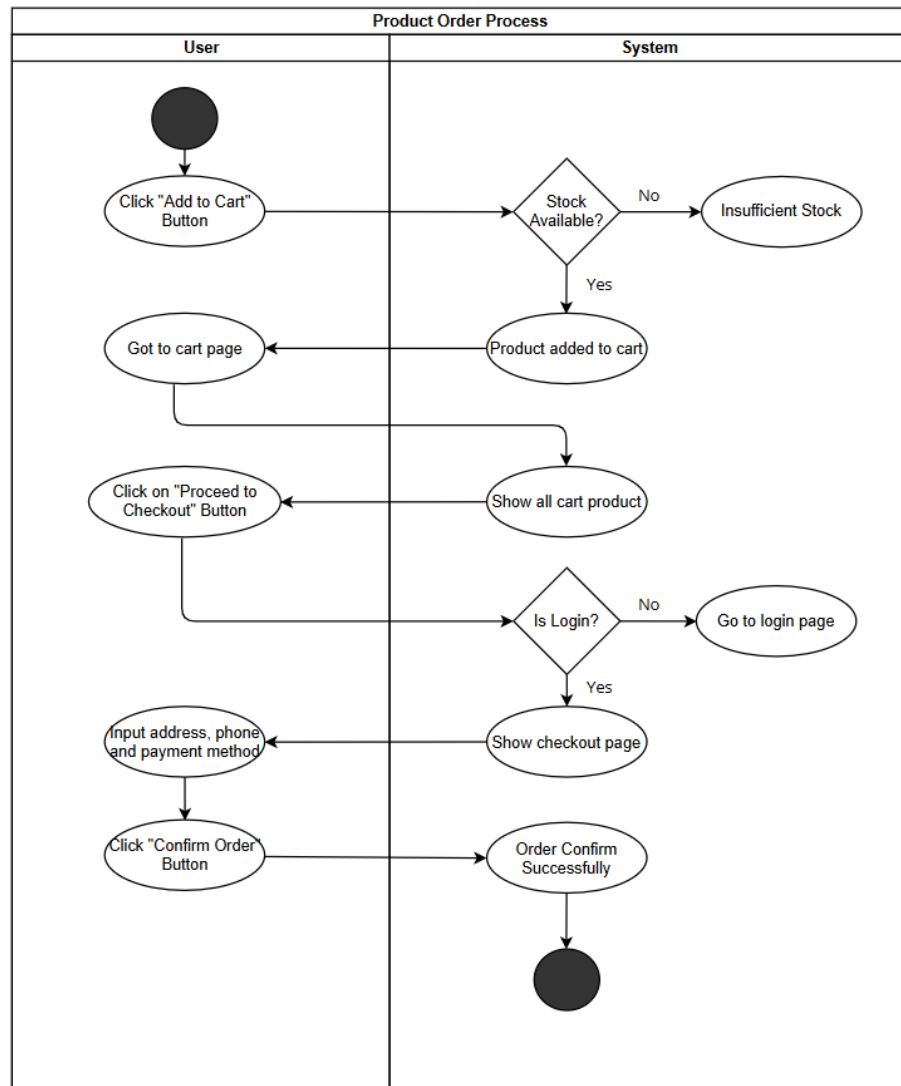


Figure 4.2.7 Product Order Process.

4.2.8. Order Management Process

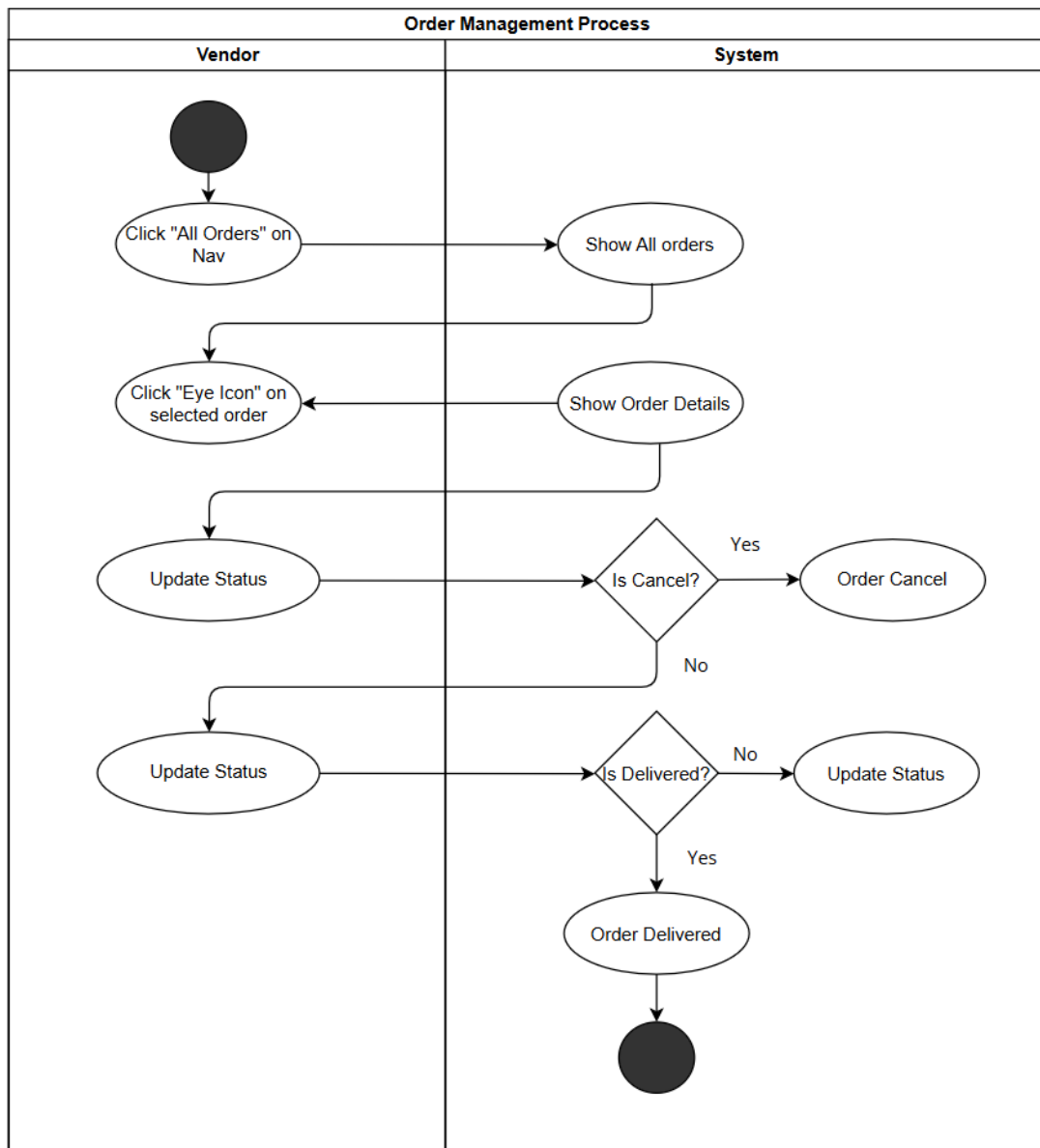


Figure 4.2.8 Order Management Process.

4.3 Sequence Diagram

A sequence diagram is a type of UML diagram that shows how objects or components interact with each other in a specific order over time. It is used to visualize the flow of messages, events, or function calls between different parts of a system, helping to understand system behavior, communication, and the sequence of operations in a process.

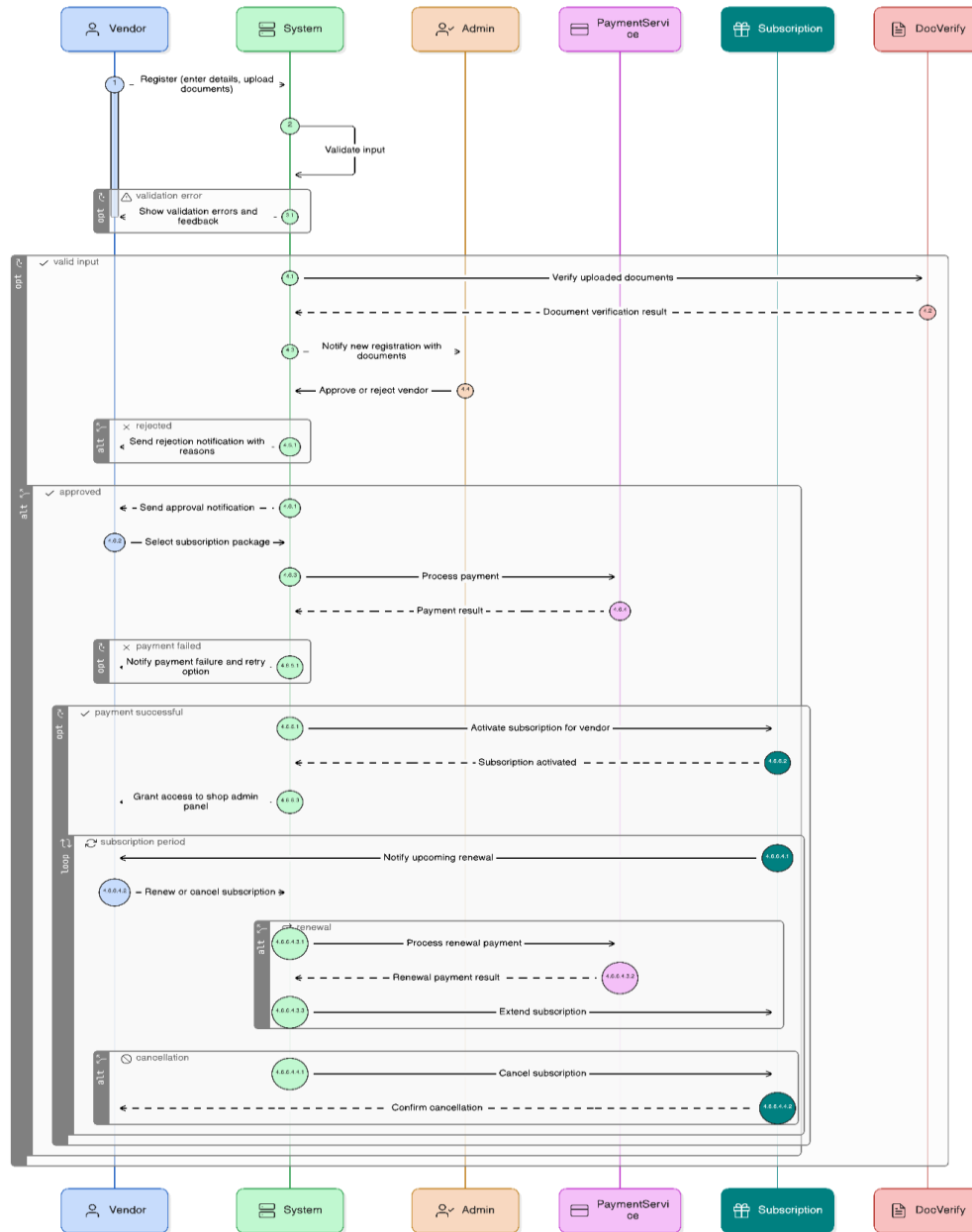


Figure 4.3 Sequence Diagram.

4.4 CRC Cards

CRC (Class-Responsibility-Collaborator) cards are a simple tool used in object-oriented design to identify and organize classes in a system. Each card represents a class and lists its responsibilities (what the class should do) and its collaborators (other classes it works with). They are mainly used during the early design phase to brainstorm, clarify roles, and visualize how different parts of the system interact. This helps teams better understand the system's structure before writing code.

Table 4.4.1 CRC of Vendor.

Class Name: Vendor	
Class Type: Role	
Class Characteristics: Tangible, Persistent, Dynamic	
Responsibilities: • Register • Login • Upload Verification Document • Manage Product, Category, Brand and Order	Collaborators: • Shop • Product • Category • Brand • Package • Admin • Order

Table 4.4.2 CRC of Admin.

Class Name: Admin	
Class Type: Role	
Class Characteristics: Tangible, Persistent, Sequential	
Responsibilities: • Vendor Approval • Package Management • Generate Report	Collaborators: • Vendor • Package

Table 4.4.3 CRC of Customer.

Class Name: User	
Class Type: Role	
Class Characteristics: Tangible, Persistent, Dynamic	
Responsibilities: • Register • Login • Browse Product • Order Product • Track Order history	Collaborators: • Product • Order • PaymentService

Table 4.4.4 CRC of Product.

Class Name: Product	
Class Type: Structure	
Class Characteristics: Tangible, Persistent, Sequential	
Responsibilities: • Store Product Details • Support CRUD	Collaborators: • Vendor • Category • Brand • Order

Table 4.4.5 CRC of Order.

Class Name: Order	
Class Type: Structure	
Class Characteristics: Tangible, Persistent, Sequential	
Responsibilities: • Store Order details • Link customer, vendor and payment	Collaborators: • Vendor • Customer • PaymentService

Table 4.4.6 CRC of SubscriptionPackage.

Class Name: SubscriptionPackage	
Class Type: Structure	
Class Characteristics: Intangible, Persistent, Sequential	
Responsibilities: • Define vendor access feature • Manage package duration and renewal	Collaborators: • Vendor • Admin • PaymentService

Table 4.4.7 CRC of PaymentService.

Class Name: PaymentService	
Class Type: Structure	
Class Characteristics: Intangible, Dynamic, Sequential	
Responsibilities: • Process payments for orders and subscriptions. • Manage package duration and renewal	Collaborators: • Vendor • Order • SubscriptionPackage • Customer

4.5 Class Diagram

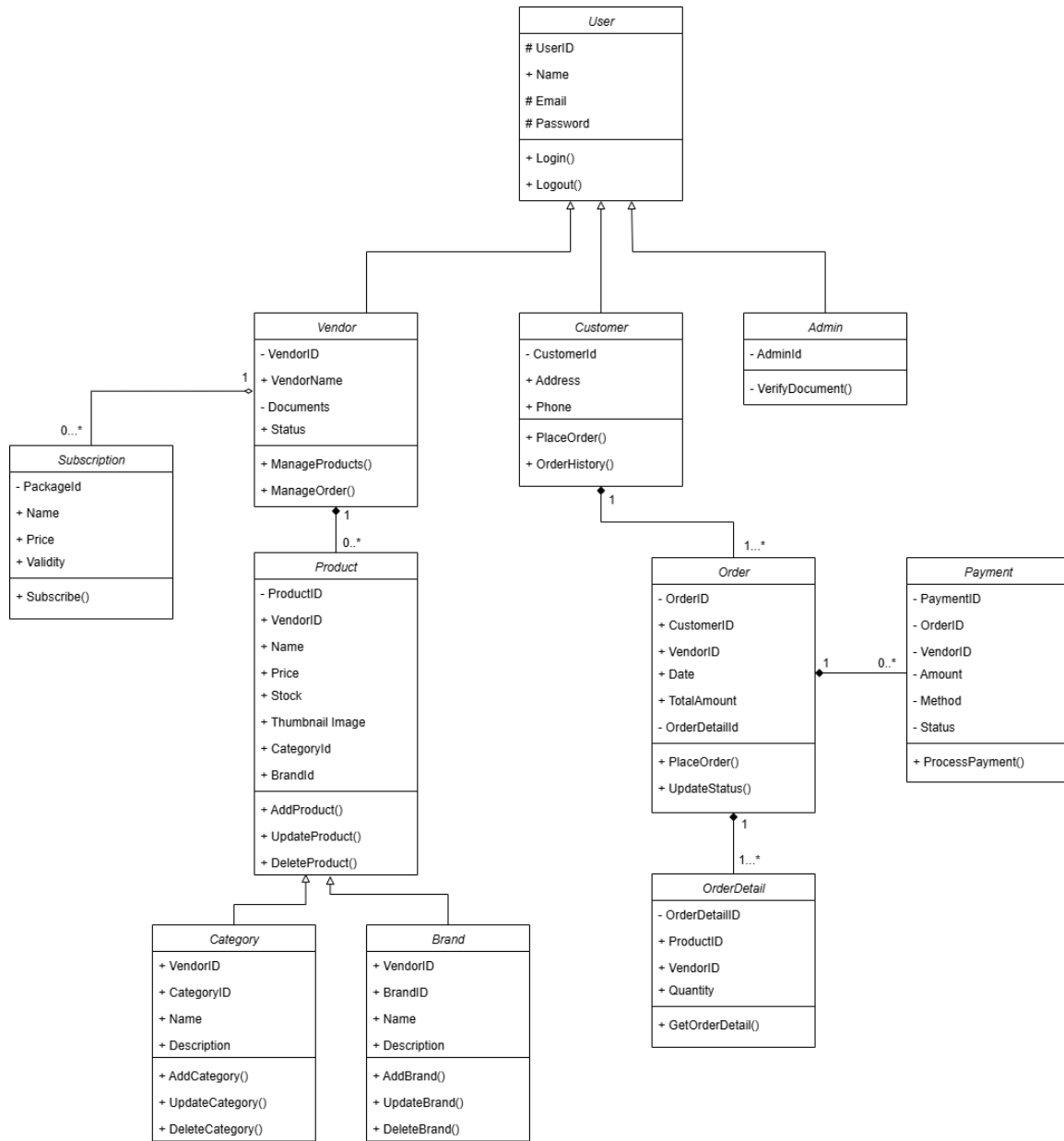


Figure 4.5.1 Class Diagram.

Chapter 5.

Project Management

Risk management in software engineering is the process of identifying, analyzing, and addressing potential issues that could negatively affect a project's success. It helps teams prepare for uncertainties like cost overruns, delays, security flaws, or technical failures. In this chapter I discussed about risk management of this project.

Stages of Risk-

- Risk Identification
- Risk Analysis
- Planning
- Monitoring

5.1 Risk Identification

Table 5.1.1 Risk Identification.

Risk Type	Possible Risk
Technical	System crashes, server downtime, or software bugs affecting vendor and customer experience.
Security	Data breaches, hacking attempts, unauthorized access, or leakage of vendor/customer data.
Financial	Payment gateway failure, fraud transactions, or incorrect subscription billing.
Operational	Vendor misuse of platform (fake products, late deliveries) causing loss of customer trust.
Legal/Compliance	Failure to comply with e-commerce laws, taxation rules, or data privacy regulations.
Project Management	Delay in project delivery, scope creep, or lack of skilled resources.
Customer Satisfaction	Poor user experience, slow response time, or lack of trust in vendors.

5.2 Risk Analysis

Table 5.2.1 Risk Analysis.

Possible Risk	Probability	Impact
System crashes, server downtime, or software bugs affecting vendor and customer experience.	30% – 40% (Moderate)	Serious
Data breaches, hacking attempts, unauthorized access, or leakage of vendor/customer data.	20% – 30% (Moderate)	Catastrophic
Payment gateway failure, fraud transactions, or incorrect subscription billing.	10% – 20% (Low)	Serious
Vendor misuse of platform (fake products, late deliveries) causing loss of customer trust.	20% – 30% (Moderate)	Tolerable
Failure to comply with e-commerce laws, taxation rules, or data privacy regulations.	10% – 15% (Low)	Serious
Delay in project delivery, scope creep, or lack of skilled resources.	20% – 30% (Moderate)	Tolerable
Poor user experience, slow response time, or lack of trust in vendors.	20% – 30% (Low)	Serious

5.3 Risk Planning

Table 5.3.1 Risk Planning.

Risk	Strategy
System downtime / crashes	Use cloud hosting with load balancing, regular backups, and monitoring tools to reduce downtime.
Security breach / hacking	Implement SSL, data encryption, firewalls, two-factor authentication, and regular security audits.
Payment gateway failure	Integrate multiple payment gateways, enable fallback methods (COD, bank transfer), and monitor transactions.
Vendor misuse (fake/late)	Set strict vendor verification, rating/review system, and penalties for poor performance.
Legal / compliance issues	Follow e-commerce laws, update terms & conditions, ensure GDPR/data privacy compliance, and consult legal experts.
Project delivery delay	Use agile methodology, set milestones, track progress, and allocate backup resources.
Low customer satisfaction	Provide 24/7 customer support, user-friendly interface, faster delivery options, and feedback mechanisms.

5.3.1. RMMM (Risk Mitigation Management and Monitoring) Planning

Table 5.3.1.1 RMMM Planning for Risk 1.

Project Risk 1	
Risk Name	System crashes, server downtime, or software bugs affecting vendor and customer experience.
Probability	30% – 40% (Moderate)

Impact	Serious
Description	System may go offline due to server failure or overload.
Mitigation in Monitoring	Use cloud hosting, load balancer, real-time monitoring tools.
Management	IT team to respond quickly, backup recovery plan in place.
Status	Managed

Table 5.3.1.2 RMMM Planning for Risk 2.

Project Risk 2	
Risk Name	Data breaches, hacking attempts, unauthorized access, or leakage of vendor/customer data.
Probability	20% – 30% (Moderate)
Impact	Catastrophic
Description	Unauthorized access or data theft could harm trust.
Mitigation in Monitoring	Ensure SSL, firewalls, two-factor login, security audits.
Management	Security team manages vulnerabilities and applies patches.
Status	Managed

Table 5.3.1.3 RMMM Planning for Risk 3.

Project Risk 3	
Risk Name	Payment gateway failure, fraud transactions, or incorrect subscription billing.
Probability	10% – 20% (Low)

Impact	Serious
Description	Payment processing errors could block sales.
Mitigation in Monitoring	Integrate multiple gateways, track payment logs, alerts for failures.
Management	Finance & IT teams to switch to backup gateway and resolve issues.
Status	Managed

Table 5.3.1.4 RMMM Planning for Risk 4.

Project Risk 4	
Risk Name	Vendor misuse of platform (fake products, late deliveries) causing loss of customer trust.
Probability	20% – 30% (Moderate)
Impact	Tolerable
Description	Vendors may sell fake products or delay orders.
Mitigation in Monitoring	Vendor verification, rating system, customer feedback monitoring.
Management	Admin to warn/suspend vendors and enforce penalties.
Status	Managed

Table 5.3.1.5 RMMM Planning for Risk 5.

Project Risk 5	
Risk Name	Failure to comply with e-commerce laws, taxation rules, or data privacy regulations.
Probability	10% – 15% (Low)

Impact	Serious
Description	Failure to follow e-commerce and data laws.
Mitigation in Monitoring	Regular legal audits, compliance checks, updated policies.
Management	Management consults legal advisors to ensure compliance.
Status	Managed

Table 5.3.1.6 RMMM Planning for Risk 6.

Project Risk 6	
Risk Name	Delay in project delivery, scope creep, or lack of skilled resources.
Probability	20% – 30% (Moderate)
Impact	Tolerable
Description	Project may miss deadlines due to scope creep.
Mitigation in Monitoring	Use agile sprints, weekly monitoring of milestones.
Management	Project manager reallocates resources and adjusts timelines.
Status	Managed

Table 5.3.1.7 RMMM Planning for Risk 7.

Project Risk 7	
Risk Name	Poor user experience, slow response time, or lack of trust in vendors.
Probability	20% – 30% (Low)
Impact	Serious

Description	Customers may leave due to poor experience.
Mitigation in Monitoring	Monitor reviews, track complaints, improve UX and support.
Management	Customer support team resolves issues and escalates feedback.
Status	Managed

Chapter 6.
Project Planning and Scheduling

Project planning and scheduling are essential activities for managing software development effectively. Project planning focuses on defining the scope, goals, resources, tasks, and risks of the project, ensuring clarity on what needs to be done and how it will be achieved. Project scheduling, on the other hand, organizes these tasks into a timeline, specifying when and in what order they should be completed. Together, they help allocate resources efficiently, track progress, manage risks, and ensure that the project is delivered on time, within budget, and according to requirements.

6.1 Function Point Estimation

Function Point Estimation is a software sizing technique used to measure the functionality of a system based on the features it provides to the user, rather than the lines of code. It helps estimate the effort, cost, and time needed for development by evaluating inputs, outputs, files, and user interactions.

6.1.1. Functionality, Input, Output

The below table includes the functionality, input, output of each function.

Table 6.1.1.1 Functionality, Input and Output.

Functionality	Input	Output
Vendor Registration	email, password	Confirmation message or error message
Vendor Verification	NID, TIN, Trade License	Confirmation message or error message
Subscription Purchase	Title, Package Name, Expire Date, Purchase Date, Status	Successfully purchase or error message
Product Management	Name, Price, Stock, Image, Status	Product added/updated/deleted in catalog

Category Management	Title, Slug, Description	Category created/updated
Brand Management	Title, Slug, Description	Brand created/updated
Customer Registration	email, password	Confirmation message or error message
Product Browsing/Search	Search query	Product list results
Order Placement	Customer selects product, payment details	Order confirmation, order ID
Payment Processing	Payment details (card, gateway, wallet)	Payment confirmation / failure notification
Admin Management	Vendor list, reports, system settings	Approval, system- wide reports, platform updates

6.1.2. Identify Complexity

Table 6.1.2.1 Identify Complexity of Data Function

Function Name	Types	Fields	DET	RET
Vendor	ILF	VendorID, Name, Email, Password, Phone, Document, Status, PackageID, Address	8	1
Customer	ILF	CustomerID, Name, Email, Password, Address, Phone, OrderHistory	7	1
Admin	ILF	AdminID, Name, Email, Password	4	1
Product	ILF	ProductID, VendorID, Name, Price, Stock, Image, CategoryID, BrandID	8	1

Category	ILF	CategoryID, Name, slug, status	4	1
Brand	ILF	BrandID, Name, slug, status	4	1
Shop	ILF	ShopID, VendorID, ShopName, Description, Vendor Name	5	1
Order	ILF	OrderID, CustomerID, VendorID, Items, TotalAmount, Status, Date	7	1
Subscription Package	ILF	PackageID, Name, Price, Duration, Features	5	1
Document Verification	ILF	TIN, NID, Trade License, VendorID	4	1

Table 6.1.2.2 Identify Complexity of Transaction Function

Function Name	Types	Fields	DET	FTR
Vendor Registration	EI	Email, Password	2	1
Vendor Verification	EO	Status	1	2
Subscription Purchase	EI	PackageID, VendorID, PaymentDetails	3	3
Product Management	EI	VendorID, Name, Price, Stock, Image, CategoryID, BrandID	7	3
Category Management	EI	Name, slug, status	3	1
Brand Management	EI	Name, slug, status	3	1
Customer Registration	EI	Email, Password	2	1
Product Browsing/Search	EQ	SearchKeywords, Filters	2	3
Order Placement	EI	OrderID, ProductID, CustomerID, VendorID, PaymentDetails	5	4
Payment Processing	EO	PaymentID, Status, Amount, Date	4	3

6.1.3. Unadjusted Function Point Contribution

Table 6.1.3.1 UFP of Transaction Function

Function Name	Types	FTR	DET	Complexity	UFP
Vendor Registration	EI	1	2	Low	3
Vendor Verification	EO	2	1	Low	4
Subscription Purchase	EI	3	3	Average	4
Product Management	EI	3	7	High	6
Category Management	EI	1	3	Low	3
Brand Management	EI	1	3	Low	3
Customer Registration	EI	1	2	Low	3
Product Browsing/Search	EQ	3	2	Low	4
Order Placement	EI	4	5	Average	4
Payment Processing	EO	3	4	Low	4
Total UFP					38

EI	1-4 DETs	5-15 DETs	16 or more DETs	EO/EQ	1-5 DETs	6-19 DETs	20 or more RETs	Complexity	Transaction Function Type	
1 FTR	Low	Low	Average	1 FTR	Low	Low	Average		EI/EQ	EO
2 FTRs	Low	Average	High	2 to 3 FTR	Low	Average	High	Low	3	4
3 or more FTRs	Average	High	High	4 or more FTRs	Average	High	High	Average	4	5
								High	6	7

Figure 6.1.3.1 Shows the UFP of Transaction Function.

Table 6.1.3.2 UFP of Data Function

Function Name	Types	DET	RET	Complexity	UFP
Vendor	ILF	8	1	Low	7
Customer	ILF	7	1	Low	7
Admin	ILF	4	1	Low	7
Product	ILF	8	1	Low	7
Category	ILF	4	1	Low	7
Brand	ILF	4	1	Low	7
Shop	ILF	5	1	Low	7
Order	ILF	7	1	Low	7
Subscription Package	ILF	5	1	Low	7
Document Verification	ILF	4	1	Low	7
Total UFP					70

ILF/EIF	1 - 19 DETs	20 - 50 DETs	51 or more RETs	Complexity	Data Function Type	
1 RET	Low	Low	Average		ILF	EIF
2 to 5 RETs	Low	Average	High	Low	7	5
6 or more RETs	Average	High	High	Average	10	7
				High	15	10

Figure 6.1.3.2 Shows the UFP of Data Function.

6.1.4. Calculation of Total Degree of Influence (TDI)

Table 6.1.4 Calculation of TDI.

SL	General System Characteristics	Brief Description	DI
01	Data Communications	System requires online communication between vendors, customers, and admin.	4
02	Distributed Data Processing	Data is processed across vendors, customers, and admin dashboards.	3
03	Performance	System must handle multiple users/vendors at the same time without slowdown.	4
04	Heavily Used Configuration	Platform will run on cloud hosting with scalability requirements.	3
05	Transaction Rate	High transaction rate due to multiple vendors, products, and customer orders.	4
06	Online Data Entry	Continuous vendor/customer interactions like registrations, orders, and updates.	5
07	End-user Efficiency	User-friendly interfaces are required for vendors and customers.	4
08	Online Update	Real-time updates for orders, inventory, and payments are required.	5
09	Complex Processing	Includes verification, subscription handling, and payment processing.	4
10	Reusability	Features like login, payment, and notifications can be reused across modules.	3
11	Installation Ease	Should be easy to deploy and configure on cloud or local servers.	2

12	Operational Ease	System should be simple to operate with minimal manual intervention.	3
13	Multiple Sites	Vendors can operate from different locations globally.	4
14	Facilitate Change	System must allow future upgrades (new features, more vendors).	4
Total TDI			52

6.1.5. Final Calculation

$$\begin{aligned}
 \text{Value Adjustment Factor (VAF)} &= 0.65 + (0.01 * \text{TDI}) \\
 &= 0.65 + (0.01 * 52) \\
 &= 1.17
 \end{aligned}$$

$$\begin{aligned}
 \text{Total UFP} &= \text{UFP (TF)} + \text{UFP (DF)} \\
 &= 38 + 70 \\
 &= 108
 \end{aligned}$$

$$\begin{aligned}
 \text{AFP (Adjustment Function Point)} &= \text{UFP (Total)} * \text{VAF} \\
 &= 108 * 1.17 \\
 &= 126.36
 \end{aligned}$$

6.2 Project Scheduling

We need to make this project successful within 12 weeks or 3 Month. So, we have created a Gantt Chart for our use in time management. Our whole timeline is shown in below Gantt Chart.

Task	Subtask	Month 1				Month 2				Month 3			
		W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Planning & Requirement Gathering	Background Study												
	Requirement Specifications												
Analysis Modeling	Use Case Diagram												
	Other Diagrams												
Design	ERD and DFD Design												
	Input, Output Design												
Coding	Coding												
	Documentation												
Testing	Self-Testing												
	Reviews												

Figure 6.2.1 Gantt Chart.

This figure 6.2.1 shows that Requirement, Analysis and Design need 6 weeks. Coding needs 3 weeks and 3 weeks.

Chapter 7.

Project Estimation

In software engineering, project cost estimation is the process of predicting the effort, time, and financial resources required to develop a software system. It involves analyzing factors such as project size, complexity, tools, and team skills to estimate overall costs. This is used to plan budgets, allocate resources effectively, set realistic timelines, and ensure the project can be delivered within financial and time constraints. The software cost estimation mostly based on:

- Personnel Cost
- Hardware Cost
- Software Cost
- Other Cost

7.1 Personnel Cost

Total days in a year = 365

Total government holidays in a year = 26

Total weekly holidays in a year = 52

Total duration in working days to finish the project = $365 - (26+52) = 287$ days

Total duration in working days per month to finish the project = $(287/12) = 23.92$ days

Daily working hours of the organization = 10

Monthly working hours of the organization = $23.92 * 10 = 239.2$ hours

Table 7.1 Personnel Salary.

Position	Salary/month	Salary/hour
System Analyst	47,480	200
Planner	35,880	150
Risk Analyzer	31,096	130
System Designer	23,920	100
Coder	19,136	80
Tester	16,744	70

Table 7.2 Personnel Cost Estimation.

Designation	Number of Person	Working Hours	Total Salary	First Payment At 60% of Salary	Remaining 40% Payment of Salary	Remaining Payment At 40% Distributed Each Week	Total Salary
System Analyst	one	717.6	1,43,520	86,112	57,408	4784	1,43,520
Planner	one	717.6	1,43,520	86,112	57,408	4784	1,43,520
Risk Analyzer	one	310.2	46,530	27,918	18,612	1551	46,530
System Designer	one	310.2	31,020	18,612	12,408	1034	31,020
Coder	one	310.2	24,816	14,889.6	9926.4	827.2	24,816
Tester	one	310.2	21,714	13,028.4	8685.6	723.8	21,714
Total							4,11,120

7.2 Hardware Cost

Table 7.2 Hardware Cost.

Item	Number	Per unit cost	Total (tk)
Coder Laptop	3	35,000	70,000
Development Server	1	2500	2500
Total Hardware Cost			72,500

7.3 Software Cost

Table 7.3 Software Cost.

Item	Cost
Visual Studio	0
PHP	0
JavaScript	0
MySQL	0
Total	0

7.4 Other Cost

Other cost refers to expenses that are not directly part of development effort (like coding or testing) but are still necessary for completing the project. These can include costs for training, travel, communication, maintenance, support, and overheads such as electricity or office space. Such costs are considered alongside development costs to get a complete and realistic project budget.

Table 7.4 Other Cost.

Category	Annual Cost (Tk)	Cost of three months (Tk)
Office Rent	1,80,000	45,000
Utility Bill	24,000	6000
Food	1,12,320	28,080
Others	1200	300
Total	3,17,520	79,380

7.5 Accounts Table

Table 7.5 Accounts Table.

Account	Amount (Tk)
Personnel	4,11,120
Hardware	72,500
Software	0
Other	79,380
Total	5,63,000

Chapter 8.

Designing

8.1 Data Flow Diagram

A **Data Flow Diagram (DFD)** is a graphical tool used in software engineering to represent how data moves through a system, showing the flow between processes, data stores, and external entities. It helps visualize how inputs are transformed into outputs and how different components of the system interact with each other. We use DFDs because they provide a clear and simple way to understand system functionality, improve communication between developers and stakeholders, identify requirements, and serve as a blueprint for designing and analyzing software systems.

8.1.1. Level 0 DFD

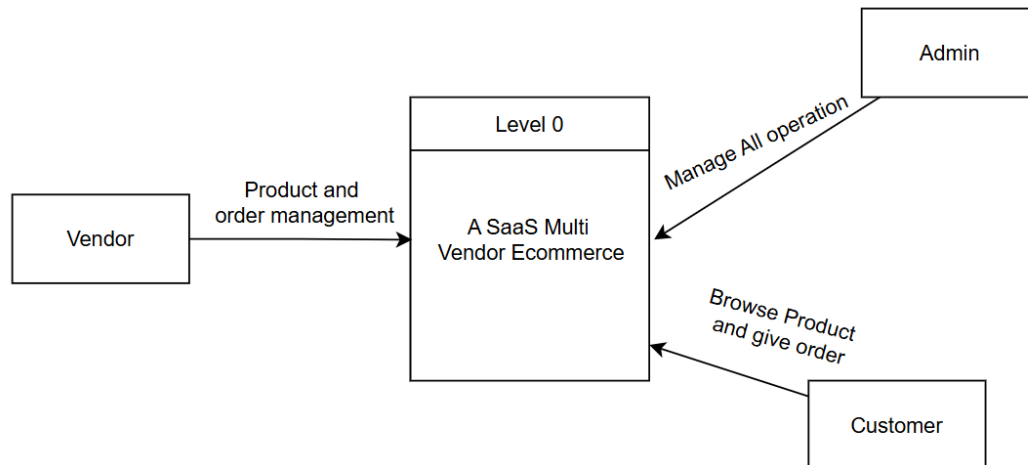


Figure 8.1.1 DFD Level 0.

8.1.2. Level 1 DFD

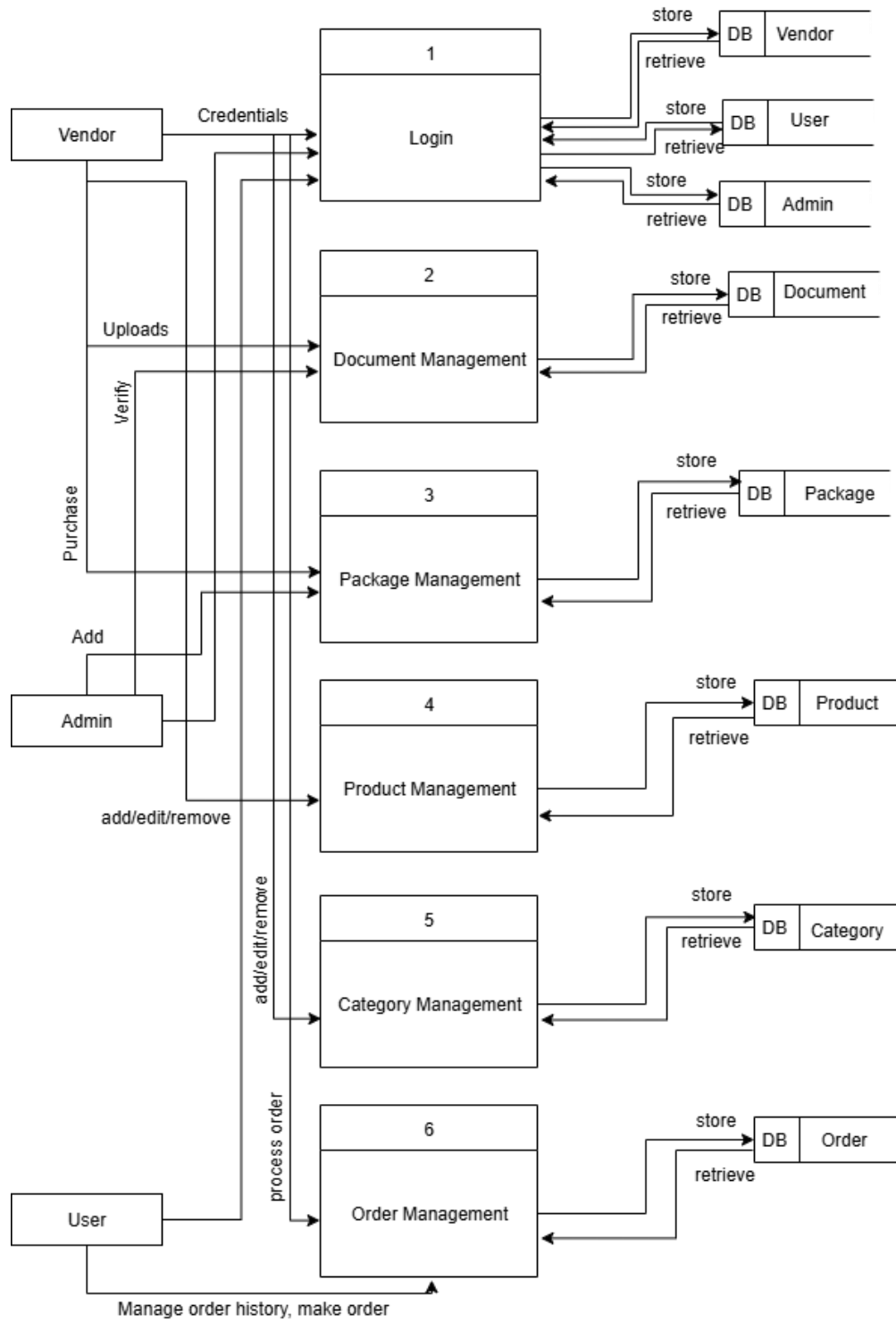


Figure 8.1.2 DFD Level 1.

8.1.3. Level 2 Process 1 DFD

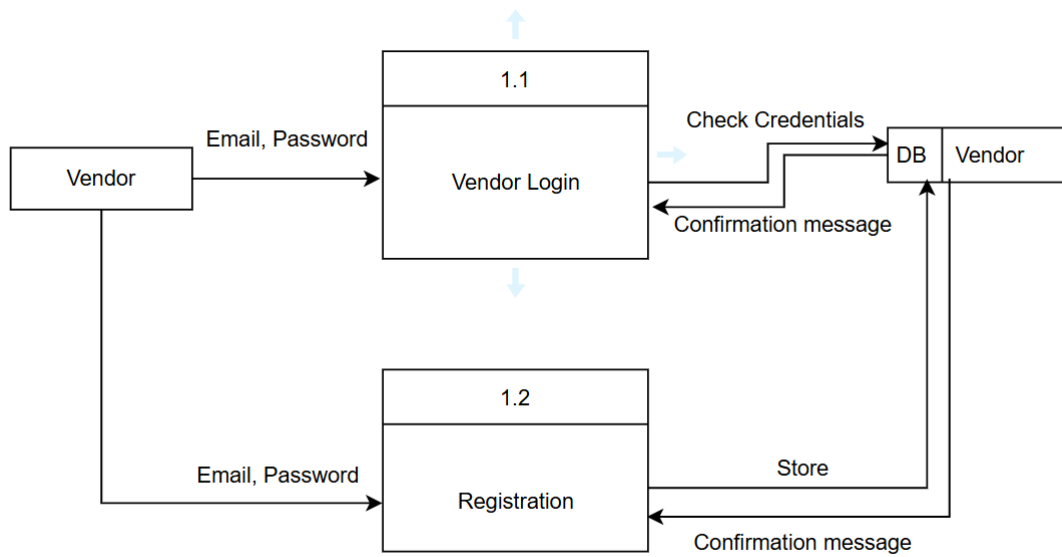


Figure 8.1.3 DFD Level 2 Process 1.

8.1.4. Level 2 Process 2 DFD

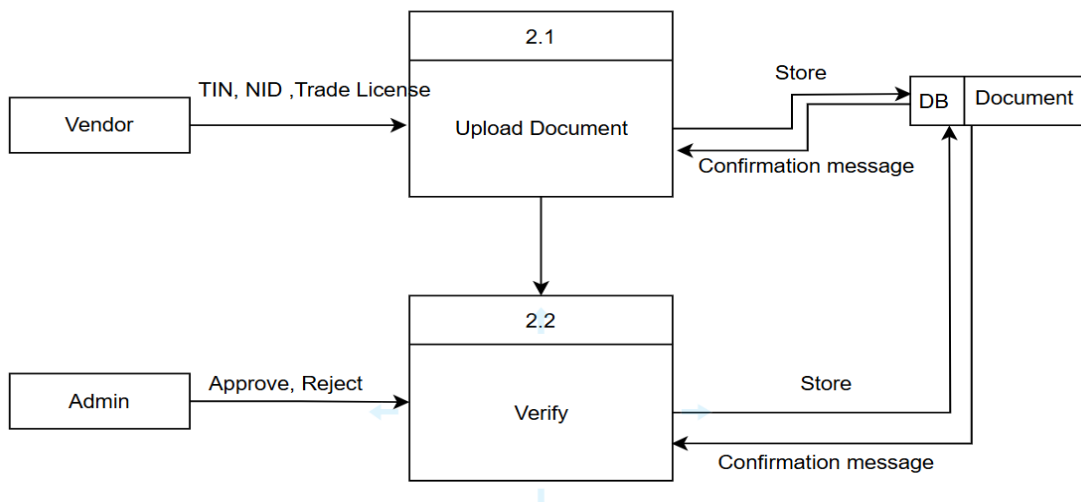


Figure 8.1.4 DFD Level 2 Process 2.

8.1.5. Level 2 Process 3 DFD

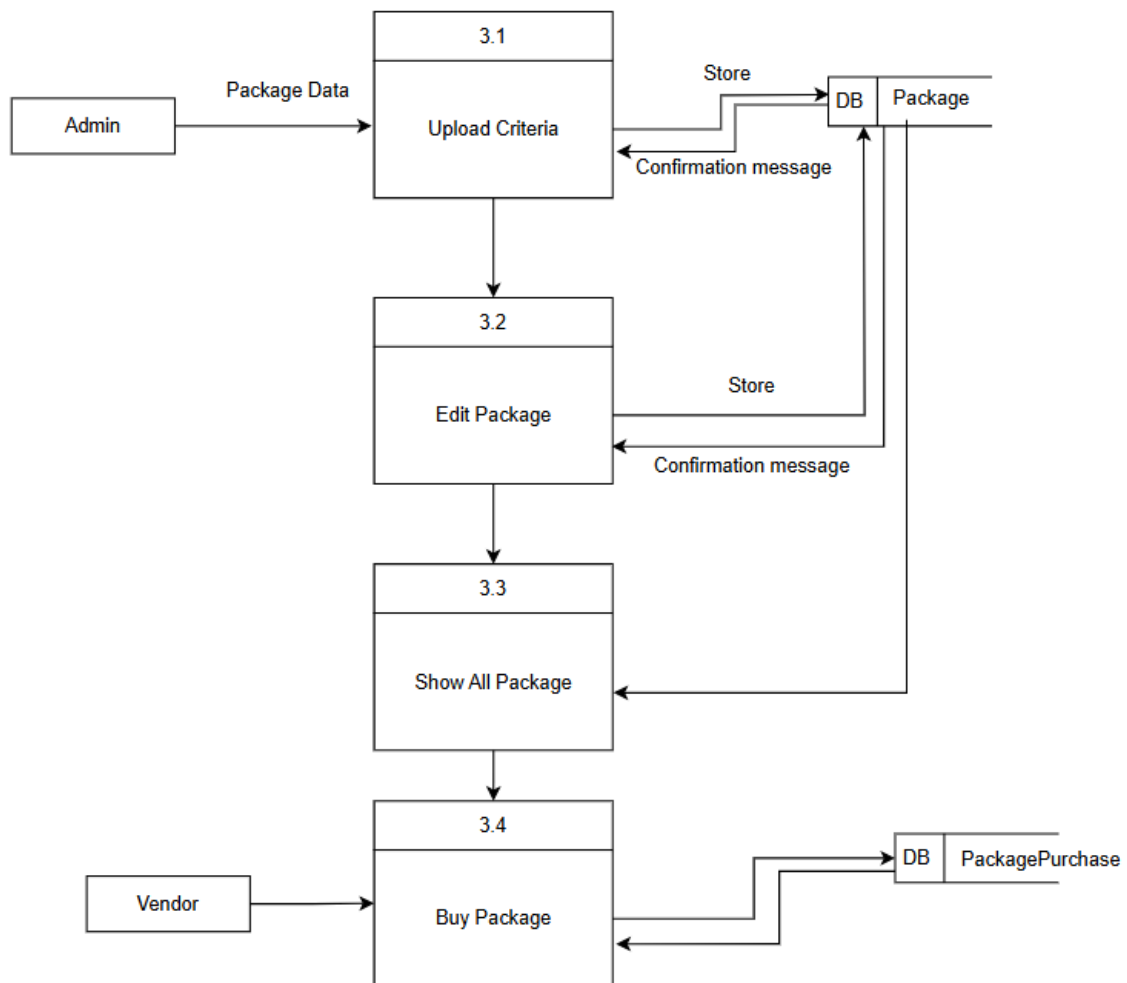


Figure 8.1.5 DFD Level 2 Process 3.

8.1.6. Level 2 Process 4 DFD

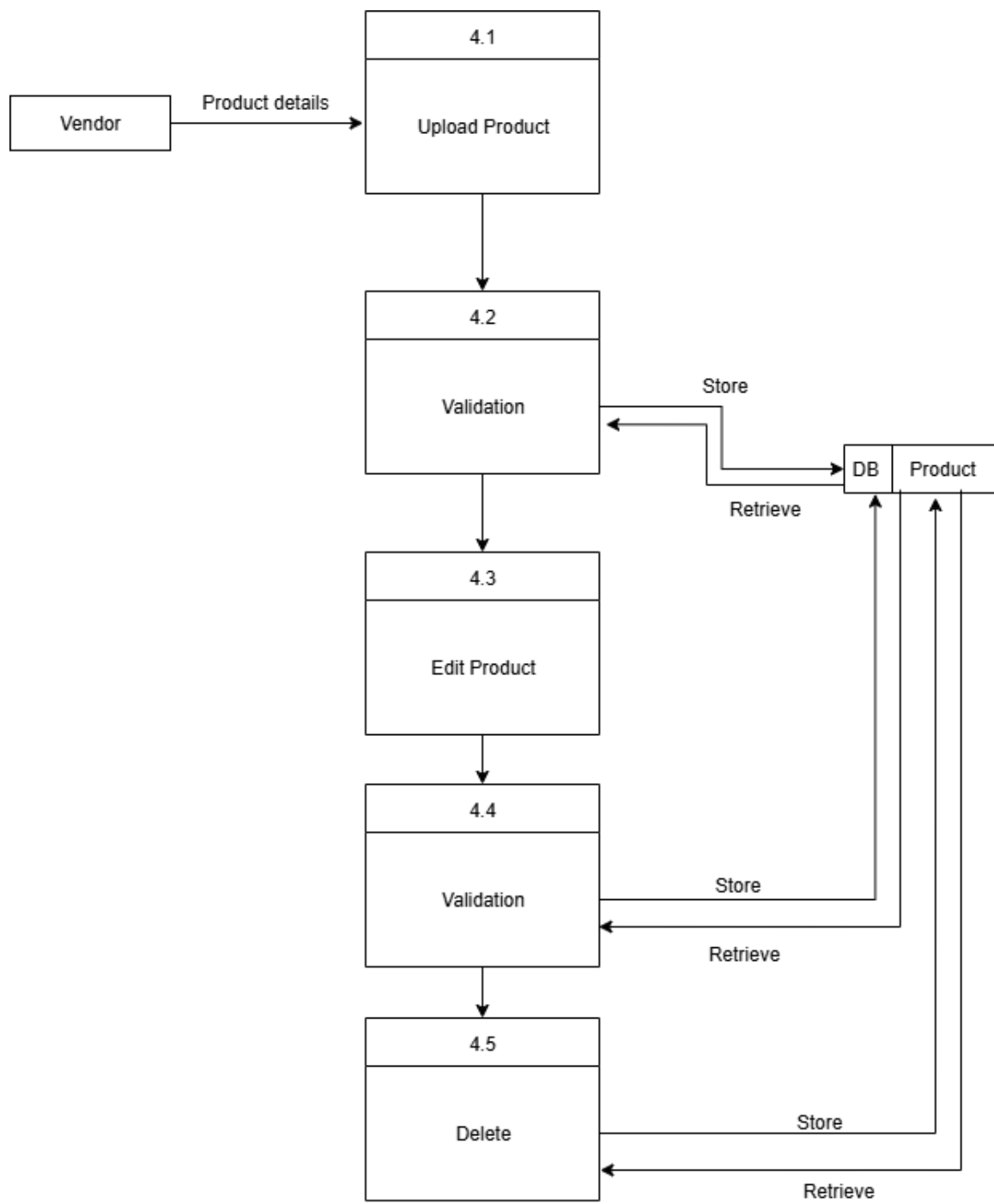


Figure 8.1.6 DFD Level 2 Process 4.

8.1.7. Level 2 Process 5 DFD

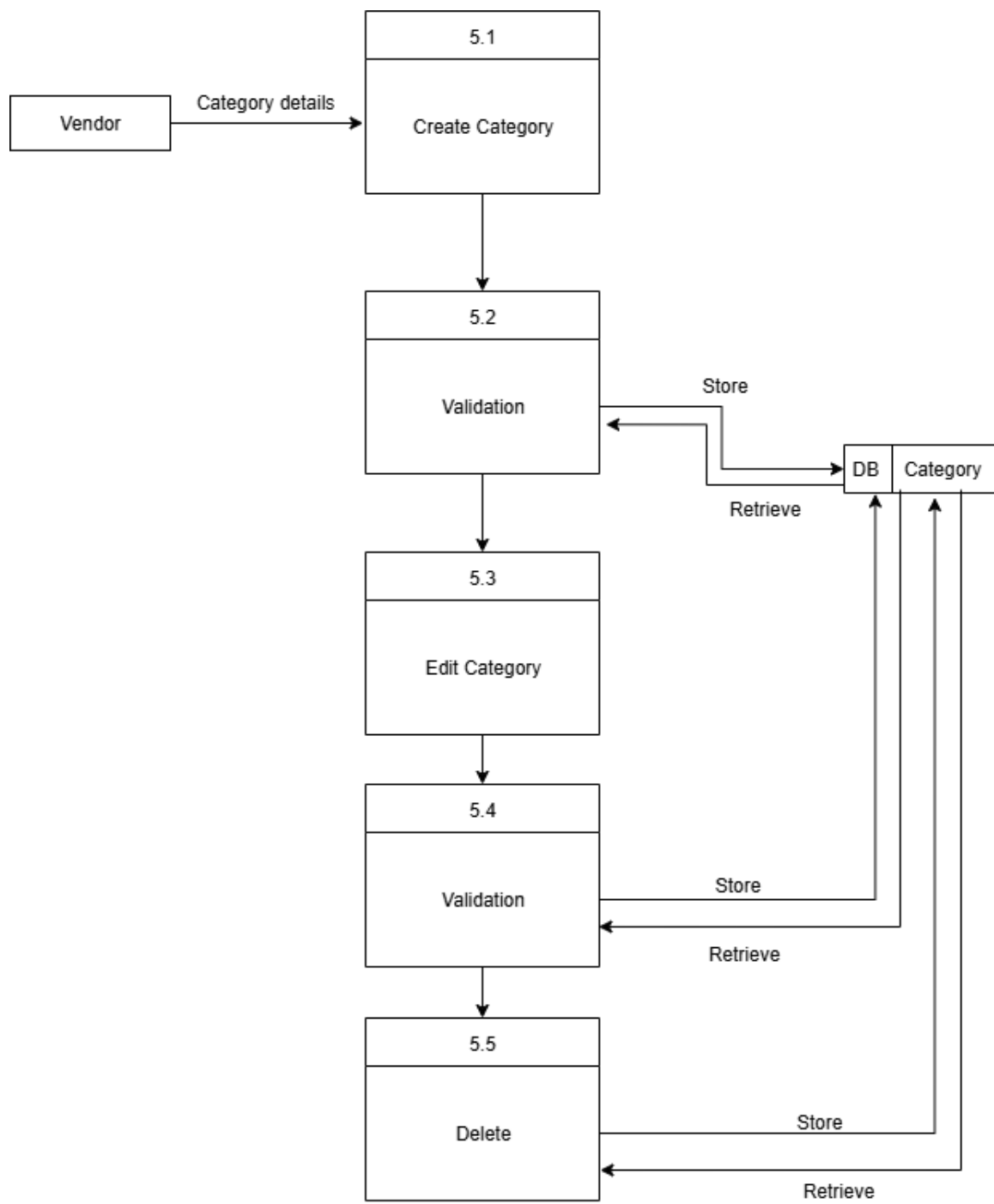


Figure 8.1.7 DFD Level 2 Process 5.

8.1.8. Level 2 Process 6 DFD

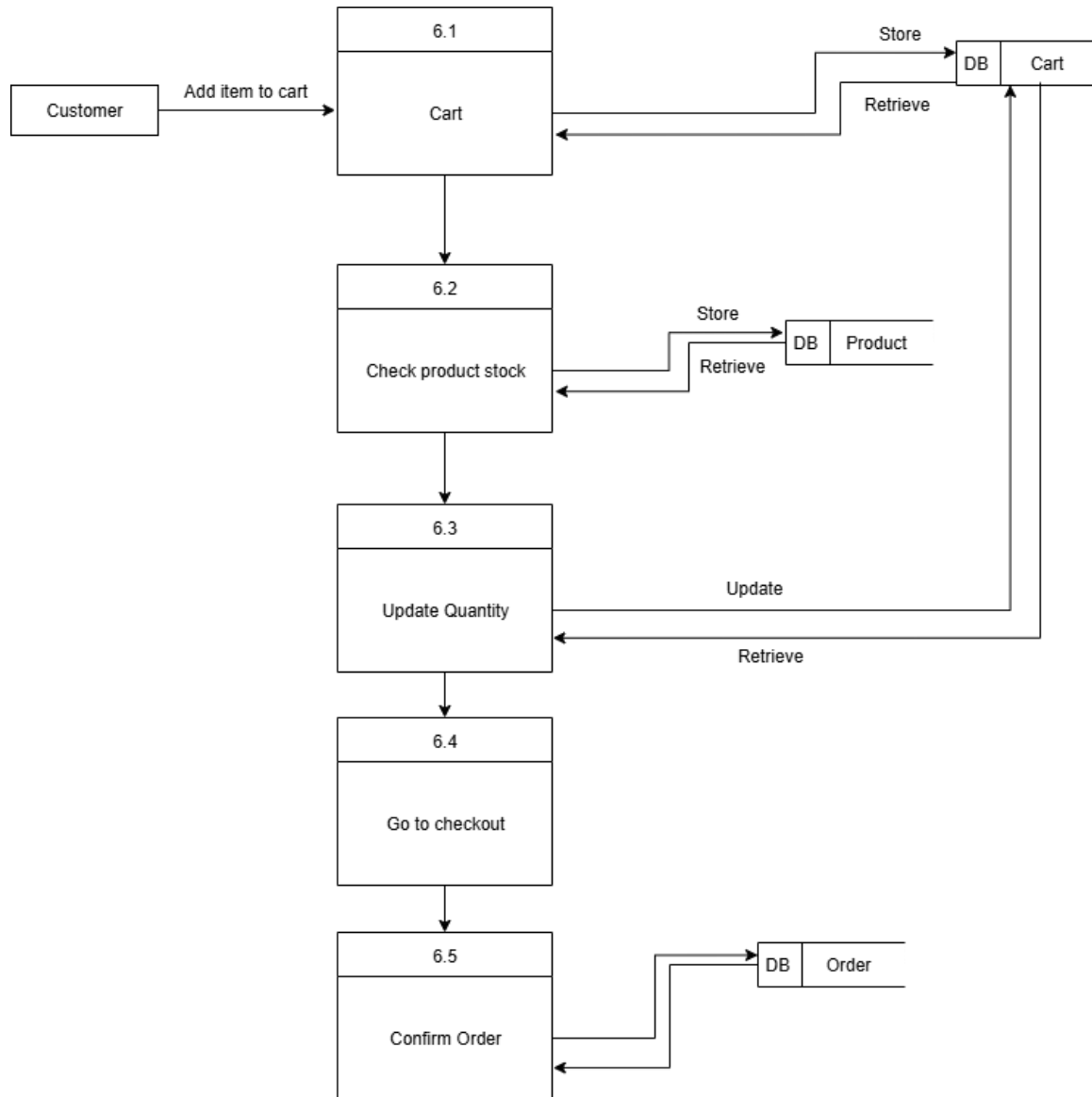


Figure 8.1.8 DFD Level 2 Process 6.

8.2 ERD Diagram

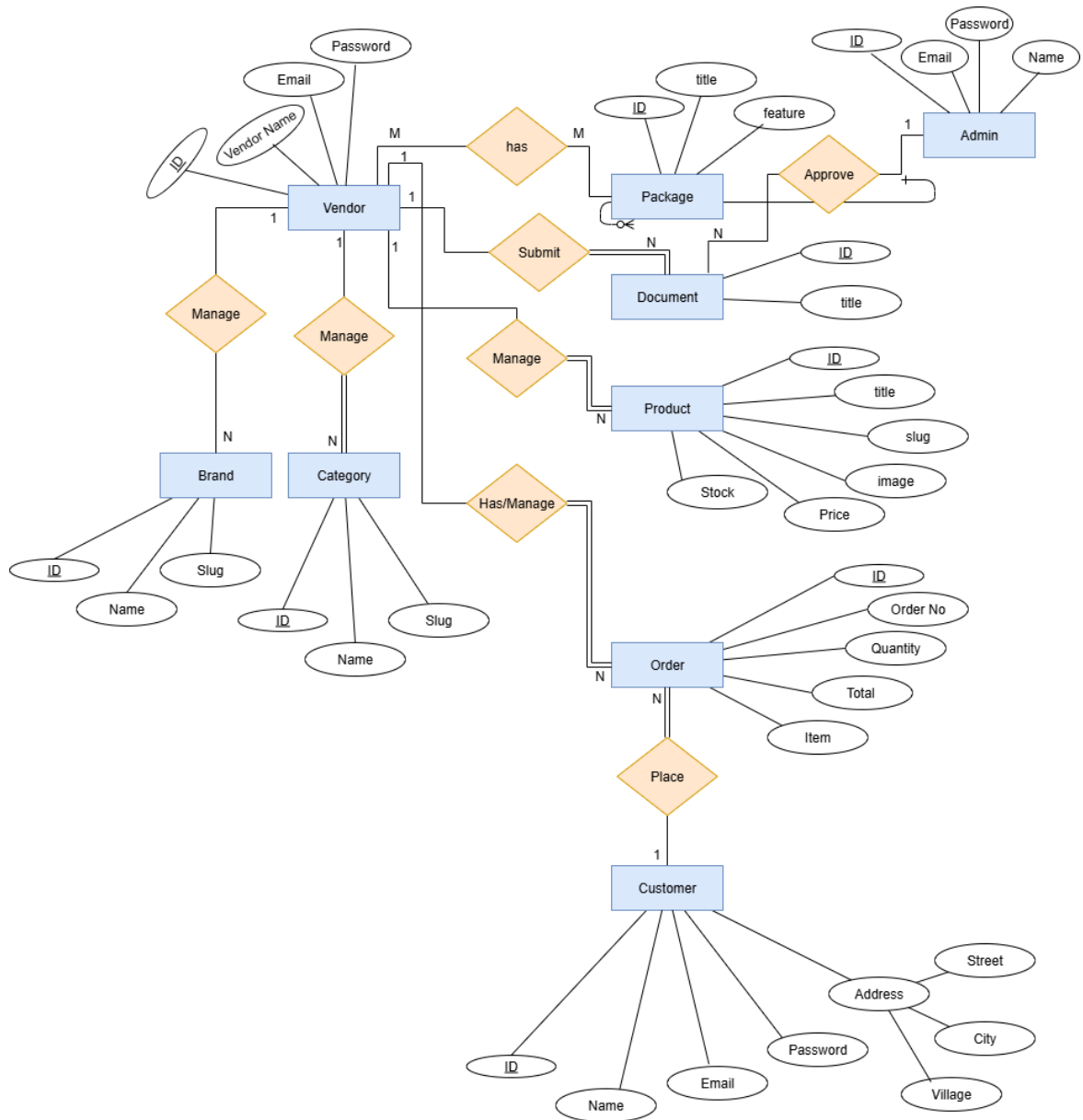
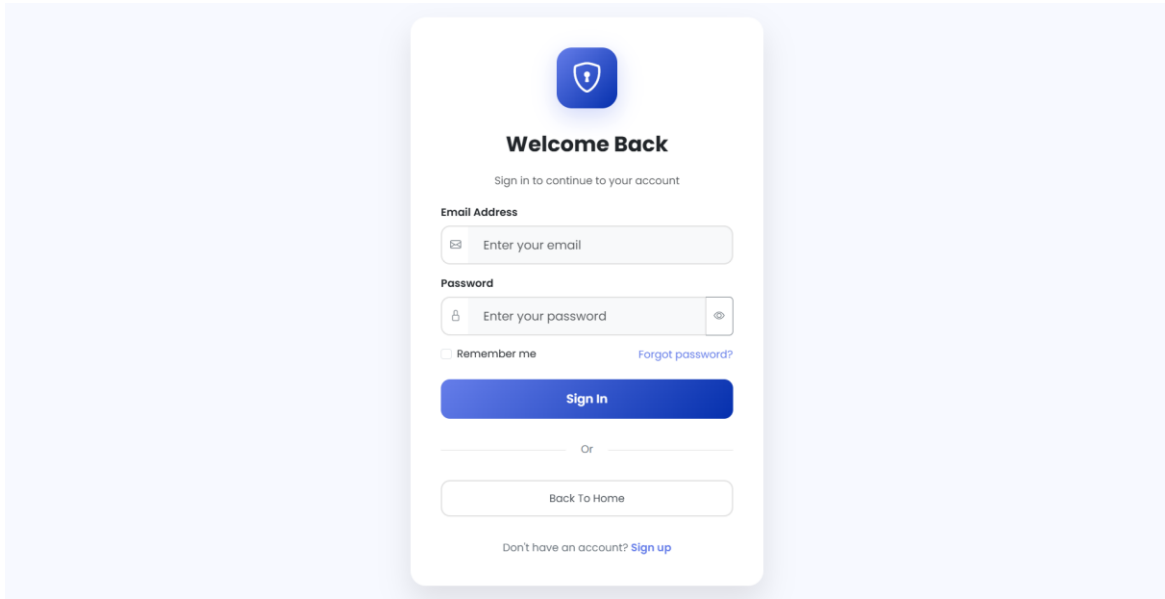


Figure 8.2 ERD Diagram.

8.3 Interface Design

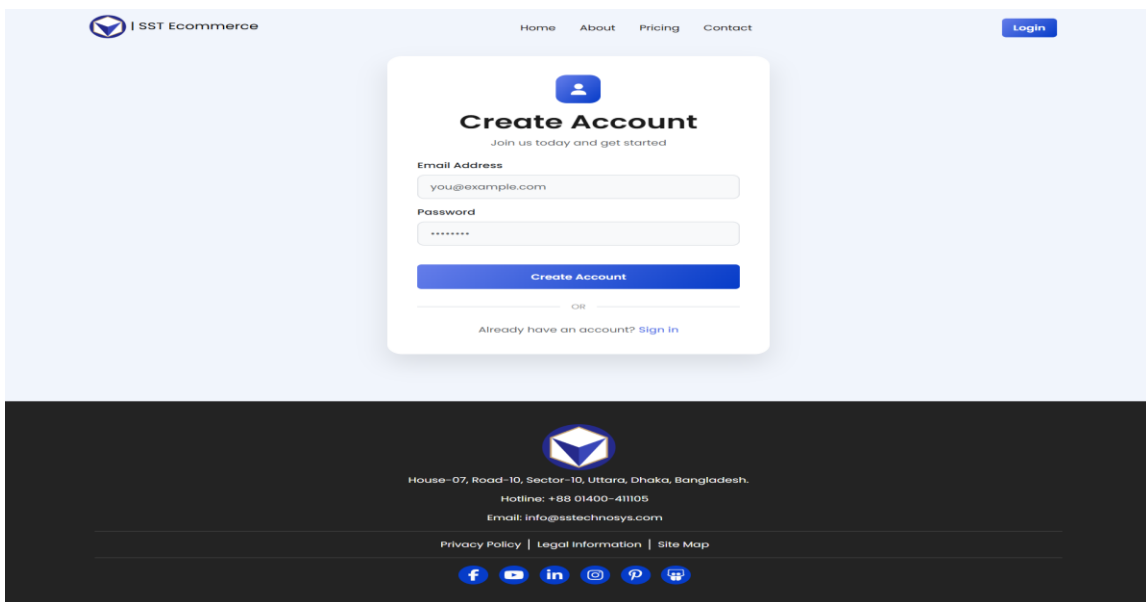
8.3.1. Vendor Login



The image shows a vendor login page with a white card centered on a light blue background. At the top of the card is a blue shield icon with a white exclamation mark. Below the icon, the text "Welcome Back" is displayed in bold, followed by "Sign in to continue to your account" in a smaller font. The form includes an "Email Address" field with a placeholder "Enter your email" and a "Password" field with a placeholder "Enter your password" and a toggle icon. Below the password field are checkboxes for "Remember me" and a link for "Forgot password?". A blue "Sign In" button is positioned below the form. Below the button is a horizontal line with the word "Or" in the center, followed by a "Back To Home" button. At the bottom of the card, there is a link "Don't have an account? Sign up".

Figure 8.3.1 Vendor Login Page.

8.3.2. Vendor Register Page



The image shows a vendor register page. At the top, there is a header with the "SST Ecommerce" logo and navigation links: "Home", "About", "Pricing", "Contact", and a "Login" button. The main content area features a white card with a blue user icon at the top. Below the icon, the text "Create Account" is displayed in bold, followed by "Join us today and get started" in a smaller font. The form includes an "Email Address" field with a placeholder "you@example.com" and a "Password" field with a placeholder "*****". A blue "Create Account" button is positioned below the form. Below the button is a horizontal line with the word "OR" in the center, followed by a link "Already have an account? Sign in". The footer is a dark blue section containing the "SST Ecommerce" logo, contact information: "House-07, Road-10, Sector-10, Uttara, Dhaka, Bangladesh.", "Hotline: +88 01400-411105", and "Email: info@sstechnosys.com". It also includes links for "Privacy Policy", "Legal Information", and "Site Map", and a row of social media icons: Facebook, YouTube, LinkedIn, Instagram, Pinterest, and Twitter.

Figure 8.3.2 Vendor Register Page.

8.3.3. Vendor Dashboard

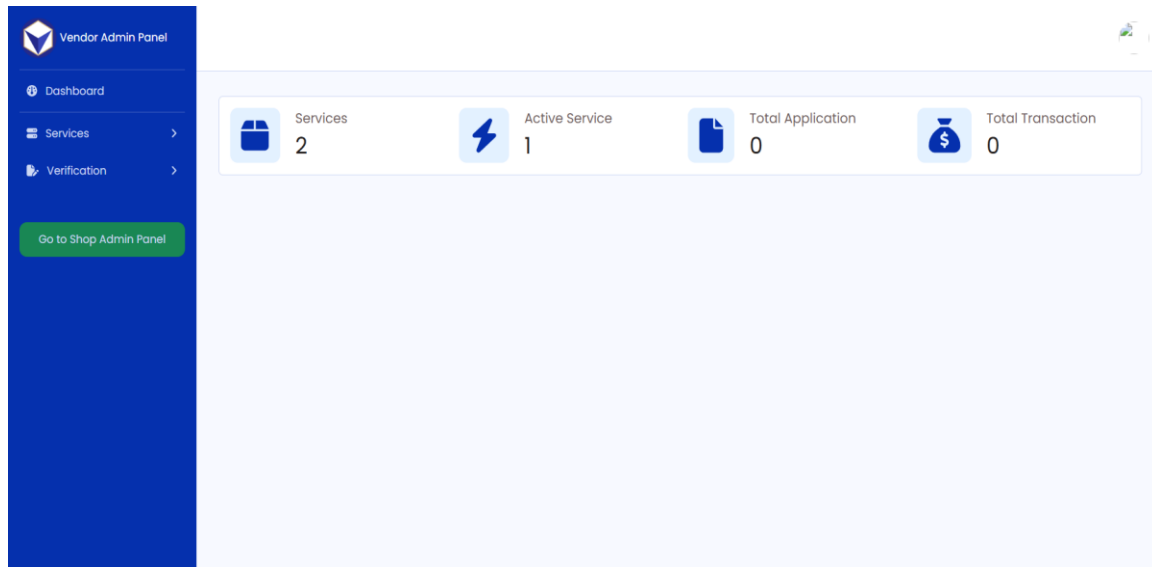


Figure 8.3.3 Vendor Dashboard.

8.3.4. Product Upload

The screenshot shows the Shop Admin Panel product upload interface. The left sidebar, titled 'Shop Admin Panel', includes a menu with 'Dashboard', 'Inventory', 'Orders', and 'Shop Setup'. A 'Go To Shop' button is located at the top of the sidebar. The main area features a breadcrumb trail 'Dashboard > ALL Products > New Product'. Below this is a 'Basic Information' section with four input fields: 'Title' (containing 'Samsung Galaxy J2'), 'Slug' (containing 'samsung-galaxy-j2'), 'Brand' (containing 'No Brand'), and 'Category' (containing 'No category'). A 'Description' field contains the text 'Best electronics product with super bass'. At the bottom, a 'Specification' table is partially visible with columns for 'Attribute' and 'Value'.

Figure 8.3.4 Product Upload.

8.3.5. Order Page

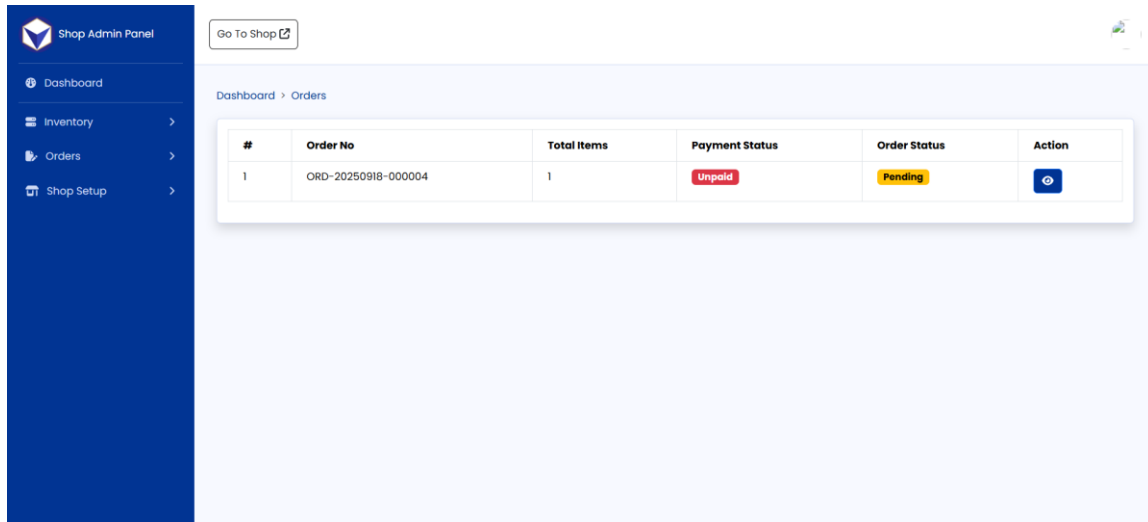


Figure 8.3.5 Order Page.

8.3.6. Order Details Page

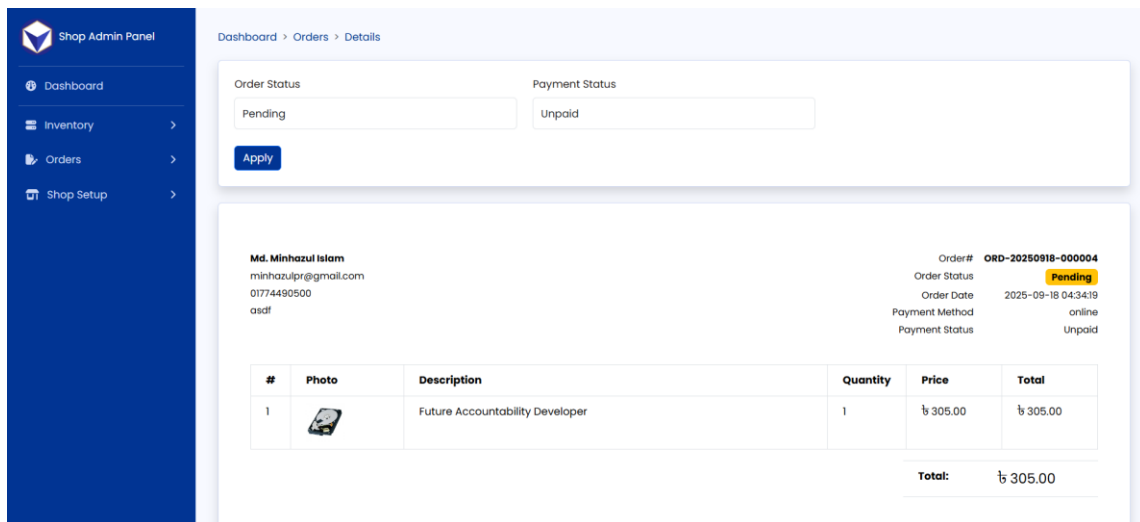


Figure 8.3.6 Order Page.

8.3.7. Shop Homepage

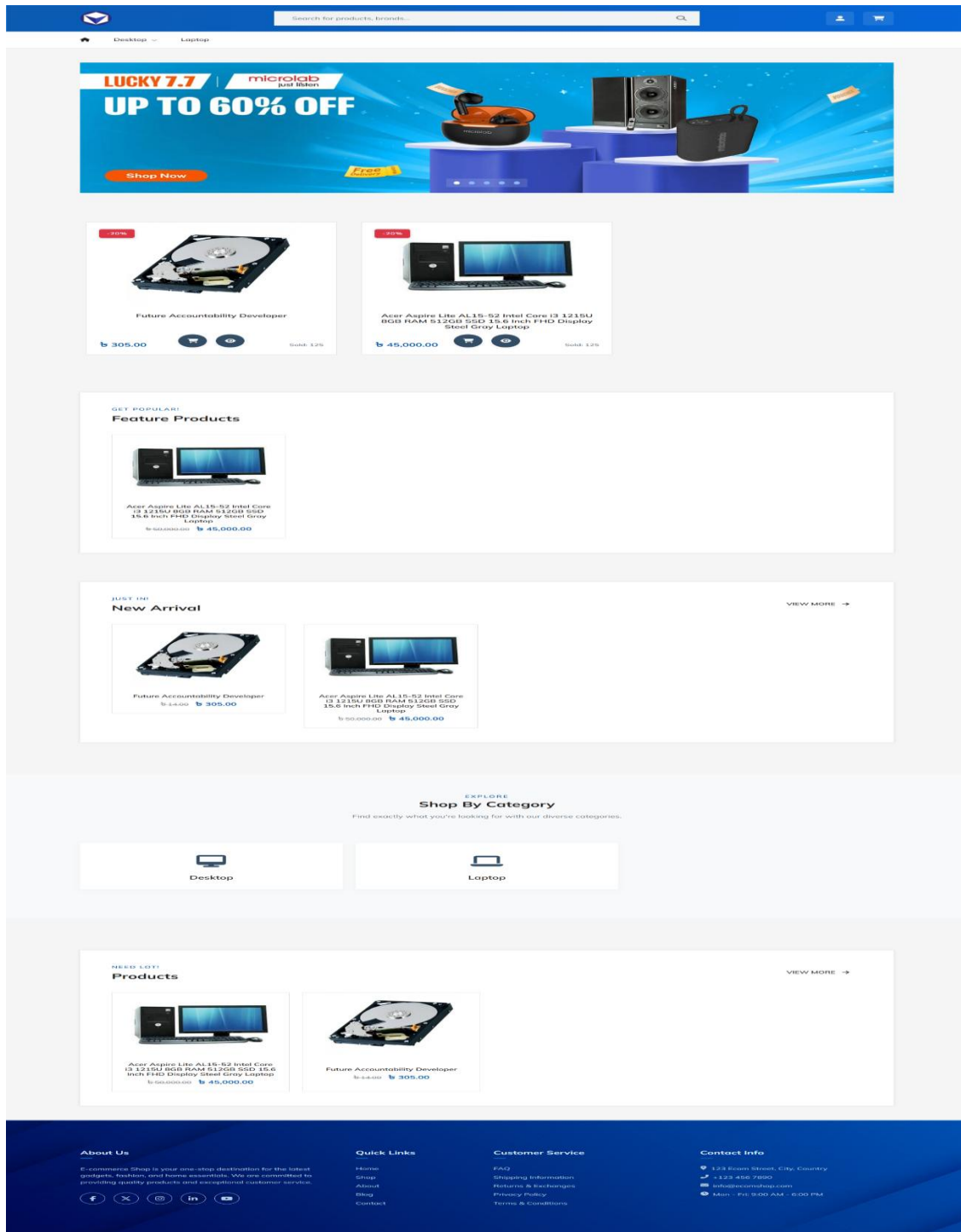


Figure 8.3.7 Shop Homepage.

8.3.8. Cart Page

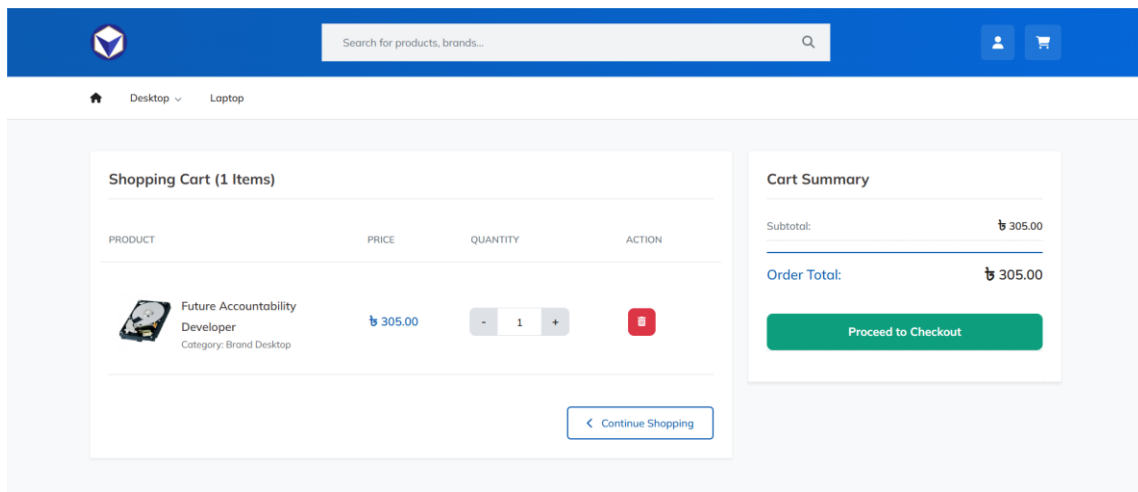


Figure 8.3.8 Cart Page.

8.3.9. Order Confirm Page

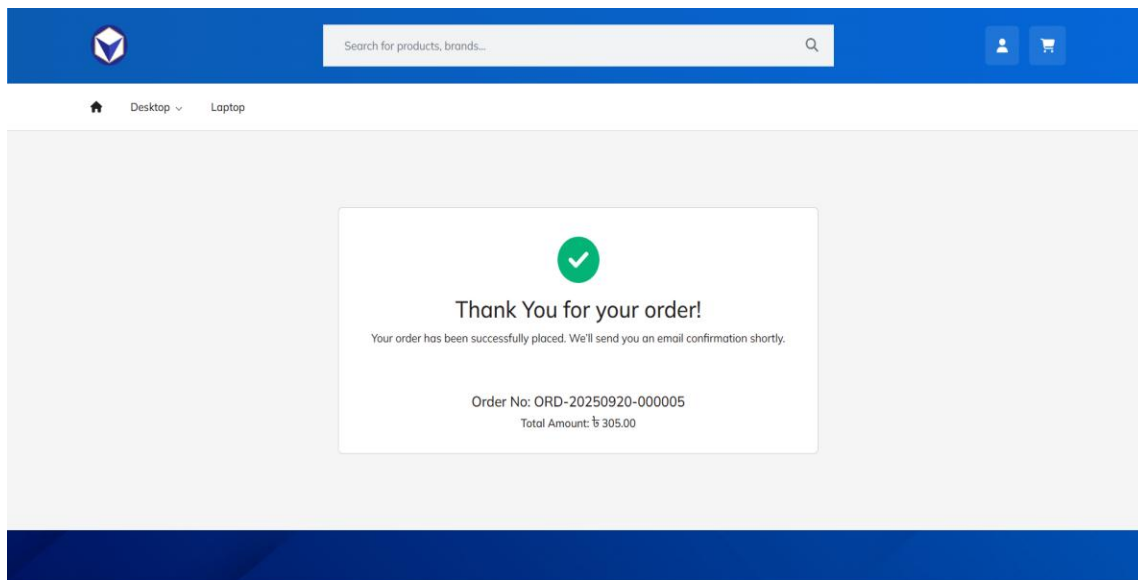


Figure 8.3.9 Order Confirm Page.

8.3.10. Admin Dashboard



Figure 8.3.10 Admin Dashboard.

Chapter 9.

Testing

9.1 Test Cases

A test case is a set of conditions or inputs designed to check whether a software feature or function works correctly. It usually includes the test data, execution steps, and the expected outcome. Test cases are used to verify that the system meets its requirements, identify defects, and ensure the software is reliable and functions as intended.

Table 9.1 Test Case 1.

Test Case ID: 1 Test Priority: High Module Name: Vendor Registration Test Title: Vendor Registration Test			Test Designed By: Md. Minhazul Islam Test Designed Date: 10-09-2025 Test Executed By: Md. Minhazul Islam			
Description: Vendor can successfully register into the system.						
Pre-Condition: Valid email, Password.						
Dependencies: none						
Steps	Test Steps	Test Data	Expected Result	Actual Result	Status	Note
1	Click “Get Started” Button	Email: vendor3@gmail.com Password: 1234	Successfully Registered and user will be redirected to dashboard	Successfully Registered and user is redirected to dashboard	Pass	
2	Enter Test Data					
3	Click on “Create Account” Button					

Table 9.2 Test Case 2.

Test Case ID: 2			Test Designed By: Md. Minhazul Islam Test Designed Date: 10-09-2025 Test Executed By: Md. Minhazul Islam			
Test Priority: High						
Module Name: User Registration						
Test Title: User Registration Test						
Description: User can successfully register into the system.						
Pre-Condition: Valid email, Password.						
Dependencies: none						
Steps	Test Steps	Test Data	Expected Result	Actual Result	Status	Note
1	Click “Register” Button	Name: Minhazul	Successfully Registered and user will get a success message	Successfully Registered and get success message	Pass	
2	Enter Test Data	Email: minhazulpr@gmail.com				
3	Click on “Register” Button	Password: 1234				

Table 9.3 Test Case 3.

Test Case ID: 3 Test Priority: High Module Name: Document Verification Test Title: Document Verification Test			Test Designed By: Md. Minhazul Islam Test Designed Date: 10-09-2025 Test Executed By: Md. Minhazul Islam			
Description: User should upload required document and admin will approve or reject.						
Pre-Condition: 1. Valid Document						
Dependencies: 1. Vendor Login, 2. Admin approval						
Steps	Test Steps	Test Data	Expected Result	Actual Result	Status	Note
1	Click “New Application” Button	NID no, NID pdf,	Document will submit and give success message	Document successfully submitted and get success message	Pass	
2	Enter Test Data	TIN no, TIN pdf,				
3	Click on “Submit” Button	Trade License no, Trade License pdf				

Table 9.4 Test Case 4.

Test Case ID: 4			Test Designed By: Md. Minhazul Islam			
Test Priority: High			Test Designed Date: 10-09-2025			
Module Name: Package Purchase			Test Executed By: Md. Minhazul Islam			
Test Title: Package Purchase Test						
Description: User should buy package and active package.						
Pre-Condition: 1. Payment Method						
Dependencies: 1. Vendor Login						
Steps	Test Steps	Test Data	Expected Result	Actual Result	Status	Note
1	Click “Subscribe” Button	None	Package will be purchased successfully and get a confirmation page	Package Purchased and get confirmation page	Pass	
2	Select Payment Method					
3	Click on “Confirm” Button					
4	Click “Services” on nav					
5	Click “Active” button					

Table 9.5 Test Case 5.

Test Case ID: 5 Test Priority: High Module Name: Add Product Test Title: Add Product Test			Test Designed By: Md. Minhazul Islam Test Designed Date: 10-09-2025 Test Executed By: Md. Minhazul Islam			
Description: User should give all product details and add product.						
Pre-Condition: none						
Dependencies: 1. Vendor Login						
Steps	Test Steps	Test Data	Expected Result	Actual Result	Status	Note
1	Click “Add New Products” on nav	Title: test product, Slug: test-product, Thumbnail Image: upload image, Price: 1000, Discount Price: 500, Stock: 10	Product will add and redirect to product list page.	Product added and redirected to product list page.	Pass	
2	Give test data					
3	Click on “Create” Button					

Table 9.6 Test Case 6.

Test Case ID: 6 Test Priority: High Module Name: Product Order Test Title: Product Order Test			Test Designed By: Md. Minhazul Islam Test Designed Date: 10-09-2025 Test Executed By: Md. Minhazul Islam			
Description: Customer can browse product and order product.						
Pre-Condition: 1. Minimum 1 product should available, 2. Stock must be sufficient						
Dependencies: 1. Product, 2. Order, 3. Payment						
Steps	Test Steps	Test Data	Expected Result	Actual Result	Status	Note
1	Browse Product		Order Confirmation Page will show with order id	Get Order Confirm Page with order id	Pass	
2	Select a product and click “Add to Cart” or “eye icon”					
3	Go to Cart Page					
4	Give item quantity	2				
5	Click “Processed to Checkout” Button	Give, Phone: 01774490500, Dhaka				
6	Select Payment method.					
7	Click “Confirm Order”					

Chapter 10.

Conclusion

10.1 Project summary

This project demonstrates the design and development of a multi-vendor e-commerce platform where vendors, admin, and customers can interact through a single, integrated system. The platform provides a structured flow: vendors register and submit documents, admin handle verification, and approved vendors can subscribe to packages and access their shop panel. From there, they manage products, categories, brands, and orders. Customers benefit by browsing products from multiple shops, placing secure orders, and tracking deliveries.

The system is technically and operationally feasible as it uses role-based access, scalable hosting, secure payments, and a user-friendly design. It addresses both business and technical requirements by ensuring trust through verification, offering flexible shop management, and enabling smooth customer experiences. Risks such as downtime, fraud, or legal compliance have been analyzed, with strategies for mitigation and monitoring.

Overall, the project provides a practical solution for enabling multiple vendors to run online shops within a unified marketplace, similar to leading platforms but tailored for simplicity and local adaptability. It not only helps vendors grow their businesses online but also offers customers variety, convenience, and trust. This ensures long-term value for all stakeholders involved.

10.2 Limitation

- **Logistics Management:** The current project does not provide a logistics management module. This means features such as tracking the movement of goods, managing delivery boys, monitoring inventory during transportation, or coordinating supply chain activities are not supported. As a result, users will need to handle logistics operations manually or rely on external systems, which may limit the efficiency and integration of the overall solution.
- **Sales Analysis:** The project lacks advanced sales analysis, so it cannot generate reports, track trends, or provide customer insights automatically. Users must analyze sales data manually, which can be time-consuming and limit data-driven decisions.

- **Real Time Notification:** The project does not include real-time notifications, so users will not receive instant updates about important events or changes. This may delay responses and reduce the system's efficiency in time-sensitive situations.
- **Customer Management:** The project lacks a dedicated customer management feature, so tracking customer details, interactions, and history must be done manually, which can reduce efficiency and affect relationship management.

10.3 Future Plan

- **Logistics Management:** Building an advanced logistics management module where vendor can manage every step of product movement and delivery man tracking.
- **Sales Analysis:** Integrate advanced sales and advertisement module to expand the current business.
- **Real Time Notification:** Implement real time communication between state holders so that without loading page data will render.
- **Integrated Chat / Customer Support:** Enable real-time communication between vendors and customers.

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